

Module-5 (a)-Lasers

Lecture ppt

Presented

By

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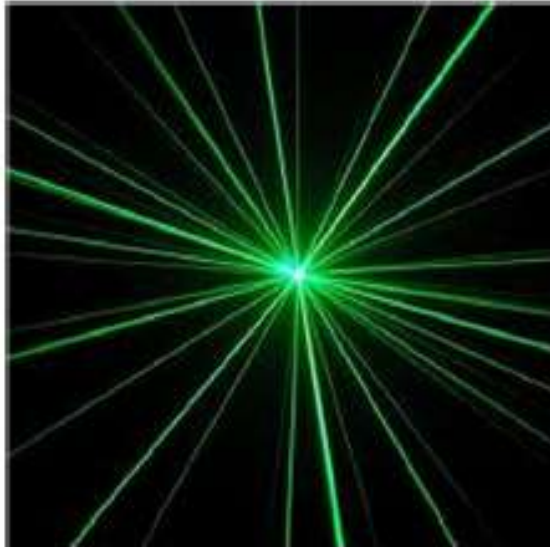
Module-5-Lasers:

Spontaneous and stimulated emission, Einstein's A and B coefficients, Population-inversion, Light amplification, Basic laser action, Ruby and He-Ne lasers, Properties and applications of laser radiation,

LASERS

1. Distinguish between spontaneous and stimulated emission. Derive the relation between transition probabilities of spontaneous and stimulated emission.
2. What is population inversion? How it is achieved in Ruby laser? Describe the construction of Ruby laser.
3. Describe with neat diagram the recording of a hologram. How does a hologram differ from an ordinary photograph?
4. What is an optical fibre? How are they classified? Define numerical aperture.
5. What are the advantages of an optical fiber? How the optical fibers are used effectively in communication
6. What are the characteristic properties of a Laser beam? Describe its important applications.
7. (a) Explain the functioning of He-Ne laser with energy level diagram. (b) How fibre optics would be used for communication?
8. Explain i) Atomic excitation ii) Transition process iii) Metastable state iv) Optical pumping.
9. Calculate the number of photons emitted per second by a 5 mW laser assuming that it emits light of wavelength $6328 \text{ \AA}$.
10. Transition occurs between a metastable state E_3 and an energy state E_2 , just above the ground state. If the emission is 1.1 micrometers and $E_2 = 0.4 \times 10^{-19} \text{ J}$. Find the energy of the E_3 state.
11. Find the numerical aperture, the acceptance angle of a step index fibre for which $n_1 = 1.5$ and $n_2 = 1.48$ and $n_0 = 1$.
12. An optical fibre has a NA of 0.20 and a cladding refractive index of 1.59. Determine the acceptance angle for the fibre in water with refractive index 1.33.
13. A pulsed laser emits light at wavelength of 694.4nm. The pulse duration is 12 ps and the energy per pulse is 0.150 J. (a) what is the length of the pulse? (b) How many photons are emitted in each pulse?

LASER



LASER Light

- short light pulses,
- spatial coherence → focusing to a tight spot over long distances

Laser Applications

- Laser Cutting
- Laser Printers
- Optical Disc Drives
- Barcode Scanners
- Laser Pointer
- Laser Surgery
- Fiber Optic
- Free-Space Communication
- Distance measurements (LUNAR LASER Ranging Experiment: precision < 4cm!!)
- many more...

A **LASER** is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation

LASER

Light Amplification by Stimulated Emission of Radiation



Before the laser

- *Before the invention of the laser the available light sources were essentially either thermal (e.g. tungsten filament lamp) or spontaneous emission from atoms and molecules (e.g. gas discharge)*
- In either case their brightness was limited by the *temperature of the emitter*
- E.g. the broadband white light of solar radiation is limited in the brightness of the spectral lines by the temperature of the gases (*recall black-body radiation*)
- Light from a *thermal or spontaneous emission source is **incoherent** – individual atoms radiating at random without any relation to one another*

Where lasers are used

- home
- factory
- hospital
- laboratory
- military
- space
- ocean
- theater

Laser light is

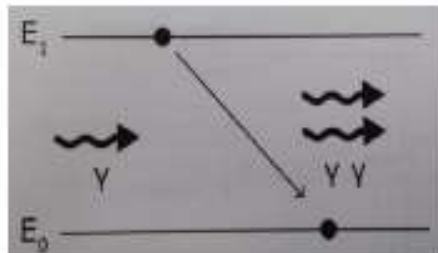
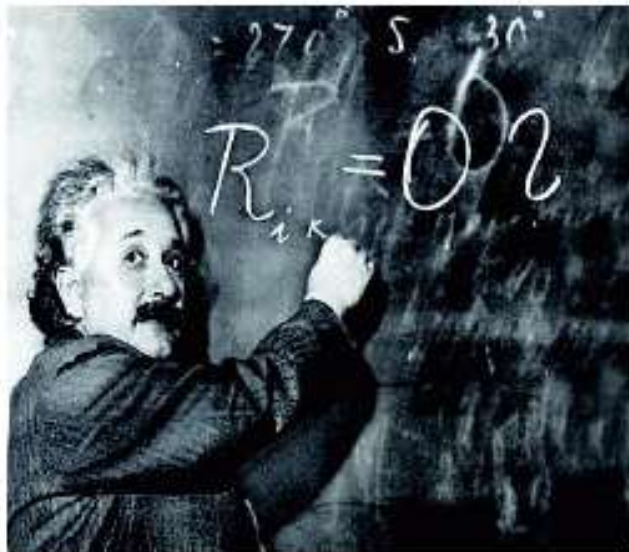
- Coherent
- Quasi-monochromatic (*nearly single frequency*), from *soft x-ray (few nm) to mid-IR (e.g. 10.2 mm)*
- Directional
- Polarized
- Can be high-power (e.g. kilo Watts)
- Can be continuously operated (*continuous wave*) or *pulsed with narrow pulse widths (picosecond, femtosecond, attosecond)*
- Can be generated from gas, liquid, solid medium (*almost anything can become a laser if you pump it hard enough!*)

HISTORICAL BACKGROUND

Discovery of Stimulated Emission in 1917

Albert Einstein

* 14.3.1879 (Ulm, Germany) † 18.4.1955, (Princeton, USA)



INVENTORS AND THEIR INVENTIONS



In 1958 Arthur L Schawlow and Charles H Townes (USA), Published a paper titled “Infrared and Optical Masers” in which it was proposed that the maser principle could be extended to the visible region of the spectrum to give rise to what later became known as a “laser” [Physical Review. 112(6), p1940, 1958]

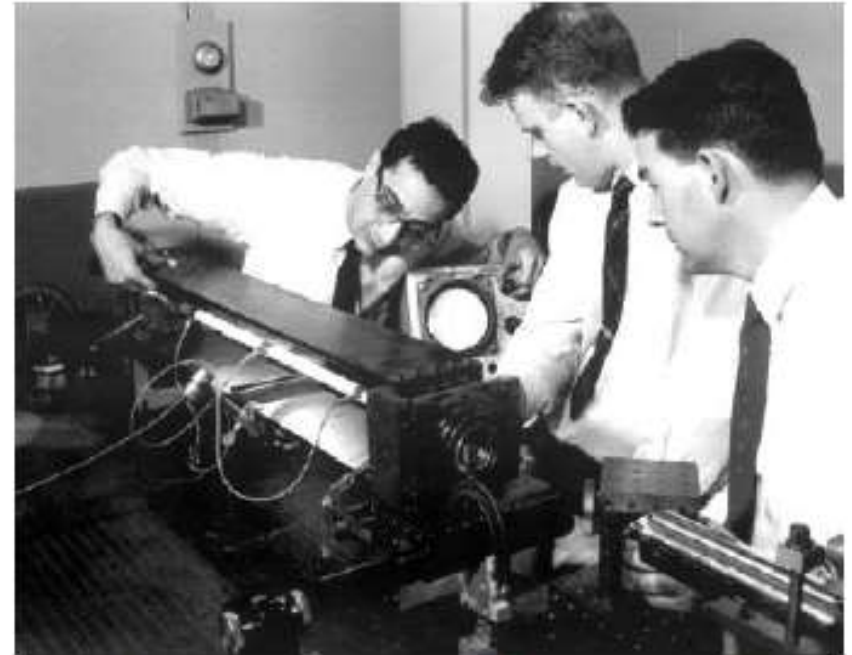


In 1960 Theodore H Maiman (USA), *Demonstrated the first laser. The laser was built at the Hughes Research Laboratories and used a rod of synthetic ruby as the lasing medium [Nature. 187, p493, 1960]*

First LASER systems: 1960

Ali Javan (*1926, Teheran/Iran)

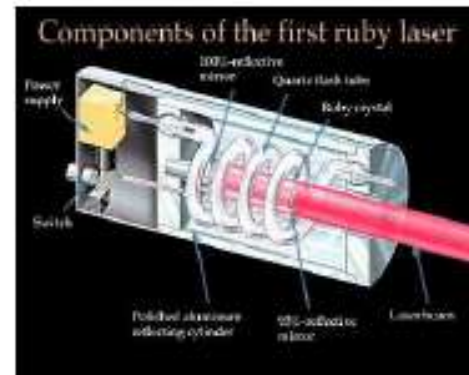
⇒ Continuous-Wave (CW) Gas
LASER



Bell Telephone Laboratories (NJ/USA)

Theodore H. Maiman (*1927, L.A./USA)

⇒ Pulsed Solid-State
LASER



Hughes Research Laboratories (CA/USA)

Nobel Prize in Physics in 1964

„...for fundamental work in the field of quantum electronics, which has led to the construction of oscillators and amplifiers based on the maser-laser principle“



Charles Hard Townes

* 28.7.1915, Greenville, USA
† 27.1.2015, Oakland, USA



Nikolay Gennadiyevich Basov

* 14.12.1922, Usman, Russia
† 1.7.2001, Moscow, Russia



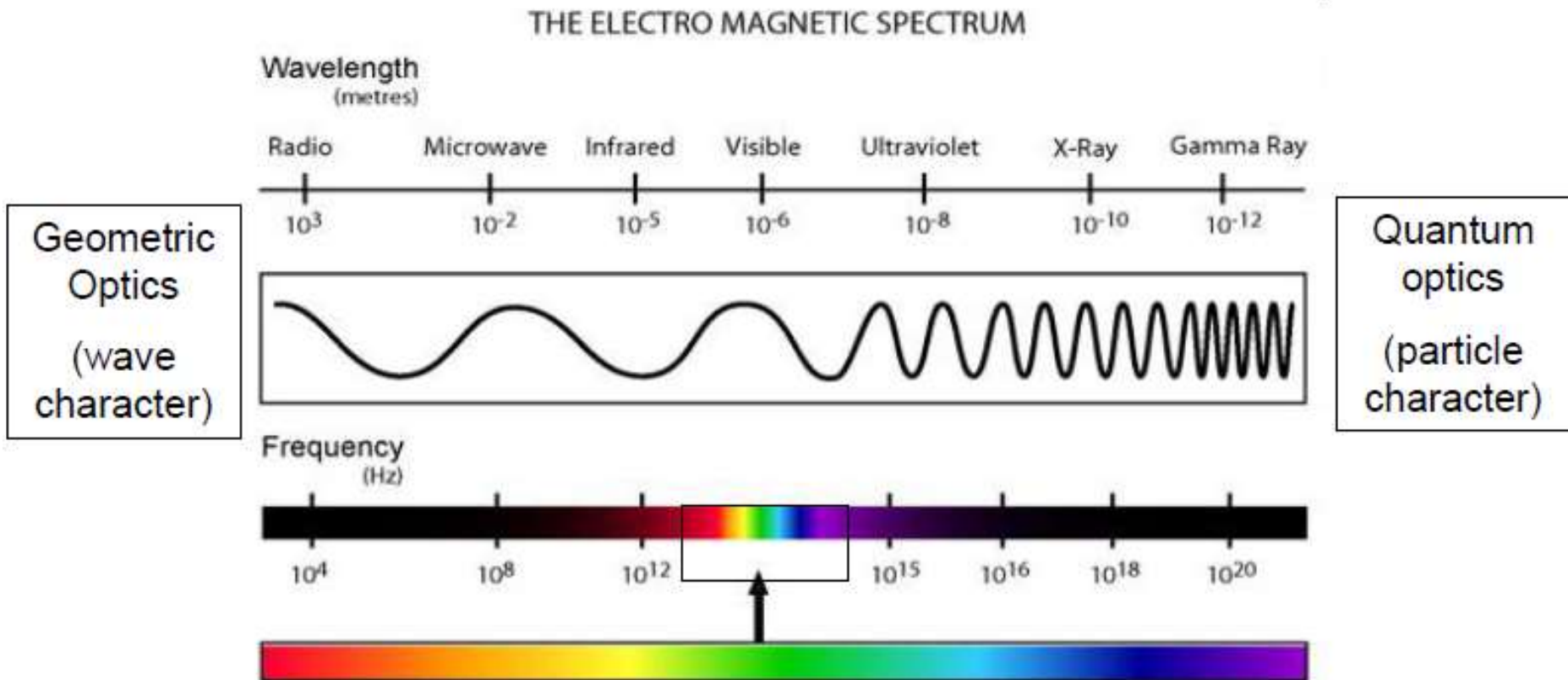
Aleksandr Mikhailovich Prokhorov

* 11.7.1916, Atherton, Australia
† 8.1.2002, Moscow, Russia

Theoretical work: MASER
principle -> LASER

Concept of optical pumping

Electromagnetic Spectrum



visible spectrum: $\lambda = 400 - 700 \text{ nm}$, $\nu = 7,5 - 4 \cdot 10^{14} \text{ Hz}$

Laser: An Introduction

Laser is an acronym for **LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION**. Laser made optical communication possible. Lasers are invented and developed between 1959 and 1962. In 1917, Einstein first predicted that the atom can emit radiation by the process of stimulated emission in addition to spontaneous emission. The idea of stimulated emission was used to construct laser by Townes and Schawlov in USA. T.H. Maiman, in 1960, constructed the first successful solid state laser, called Ruby laser. The gas laser, He-Ne laser, was developed by Ali Javan and his co-workers in 1961. In 1962, solid state diode laser was developed by Hall and his associates.

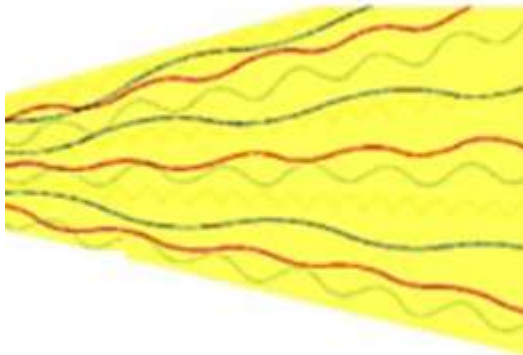
Outline History of the Development of the Laser

The historical development of Laser can be tabulated as follows:

<i>Year</i>	<i>Name</i>	<i>Achievement</i>
1916	Albert Einstein	Theory of light emission. Concept of stimulated emission.
1928	R.W. Landenburg	Confirmed existence of stimulated emission.
1940	V.A. Fabrikant	Noted possibility of population inversion.
1947	W.E. Lamb and R.C. Rutherford	First demonstration of stimulated emission.
1951	Charles H. Townes and Schawlov	The inventor of the MASER (Microwave Amplification of Stimulated Emission of Radiation).
1960	Theodore Maiman	Invented first working LASER based on Ruby. May 16th 1960, Hughes Research Laboratories.
1961	Ali Javan, William Bennet and Donald Herriot Jr.	Invented Helium Neon (He-Ne) LASER at Bell Labs.
1962	Robert Hall	Invention of semi-conductor LASER at General Electric Labs.

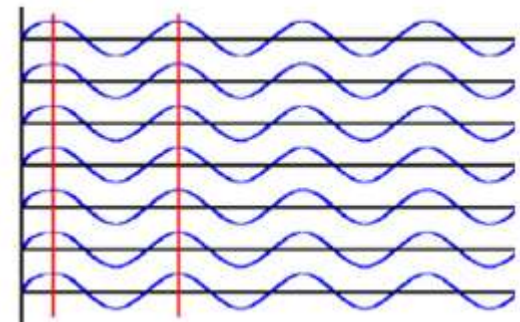
Comparison of Ordinary Beam of Light and Laser Beam

<i>S. No.</i>	<i>Ordinary beam</i>	<i>Laser beam</i>
1	It is not monochromatic.	It is monochromatic.
2	It is incoherent, i.e. the constituent waves are generally not in the same phase.	It is coherent, i.e. the constituent waves are exactly in the same phase.
3	It does not travel as a concentrated and parallel beam.	It travels as a concentrated parallel beam.
4	Ordinary light is produced by spontaneous emission.	Laser beam is produced by stimulated beam.



Out put of Common light source

➤ Light beams of random phase varying with time and position



Out put of Laser source

➤ Light of in step waves of identical frequency, phase and polarization

Interaction of Radiation with Matter

A material medium is composed of identical atoms or molecules, each of which is characterized by a set of discrete allowed energy states. An atom can move from one energy state to another when it receives or releases an amount of energy equal to the energy difference between those two states. It is termed as a quantum transition.

Let us consider two energy states E_1 and E_2 of an atom. E_1 is the lower energy state while E_2 is the excited state. As the constituent atoms of the medium are identical, the energy states E_1 and E_2 will be common to all atoms in the medium.

Let a monochromatic radiation of frequency ν be incident on the medium. The radiation may be viewed as a stream of photons, each photon carrying an energy $h\nu$. If $h\nu = E_2 - E_1$, the interaction of radiation with atoms leads to the following three distinct competing processes in the medium.

- (a) Stimulated absorption
- (b) Spontaneous emission
- (c) Stimulated emission

(a) Stimulated absorption

An atom residing in the lower energy state E_1 may absorb the incident photon and jump to the excited state E_2 . The transition is known as stimulated absorption or induced absorption (see Figure 7.1).

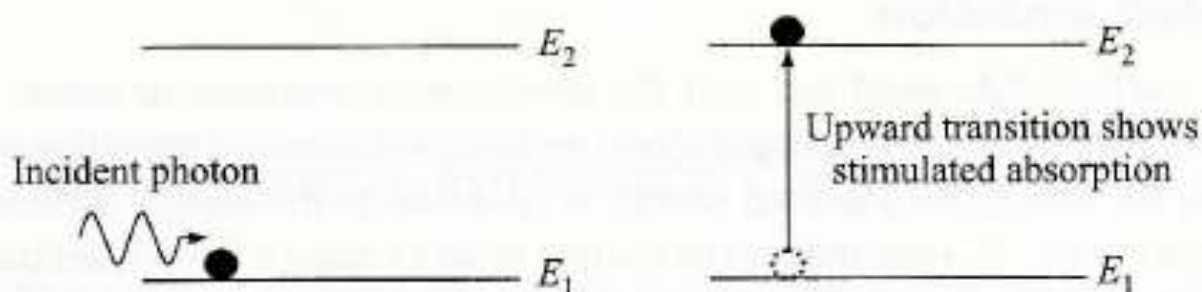
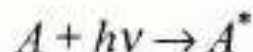


Figure 7.1 Stimulated absorption.

Corresponding to each transition made by an atom one photon disappears from the incident beam. It may be represented as



where A is the atom in lower state and A^* is an excited atom.

The probable rate of transition from $1 \rightarrow 2$ (State 1 to State 2) depends on the properties of States 1 and 2 and is proportional to the energy density $u(\nu)$ of the incident radiation of frequency ν .

$$P_{12} = B_{12} u(\nu)$$

B_{12} is a constant of proportionality, which depends on the properties of states 1 and 2. It is called Einstein's coefficient for absorption of radiation.

(b) Spontaneous emission

When the excited atom in the state E_2 returns to the lower state E_1 naturally, due to the tendency to attain minimum potential energy. During the transition the excess energy is released as a photon of energy $h\nu = E_2 - E_1$. This type of process in which photon emission occurs without any interaction with external radiation is called spontaneous emission (see Figure 7.2).

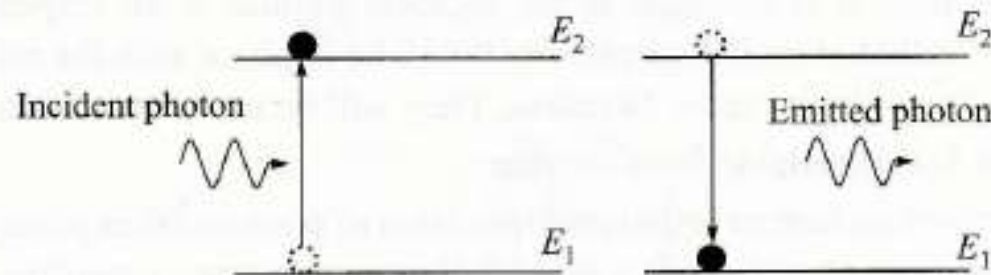
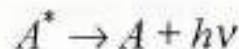


Figure 7.2 Spontaneous emission.

It may be represented as



The probability of spontaneous emission $2 \rightarrow 1$ depends only on the properties of State 1 and State 2.

According to Einstein the probable rate of spontaneous emission is denoted by

$$(P_{21})_{\text{spontaneous}} = A_{21}$$

A_{21} is called Einstein's coefficient of spontaneous emission of radiation.

Note: The mean life time τ for spontaneous emission by excited atoms is of the order of 10^{-8} seconds. However, there are some states for which τ is much longer, perhaps 10^{-3} seconds. These are called meta-stable states. These states play a major role in laser operation.

(c) Stimulated emission

An atom in the excited state need not wait for spontaneous emission to occur. There exists an alternate mechanism by which an excited atom can make a downward transition and emit light. An incoming photon for which the quantum energy $h\nu$ is equal to the energy difference between its present level (i.e. $h\nu = E_2 - E_1$) can induce the excited atom to make a downward transition realising the energy in the form of a photon. Thus, the interaction of a photon with an excited atom triggers the excited atom to drop to the lowest energy state giving up a photon. The phenomena of forced emission of photons are called induced emission or stimulated emission (see Figure 7.3).

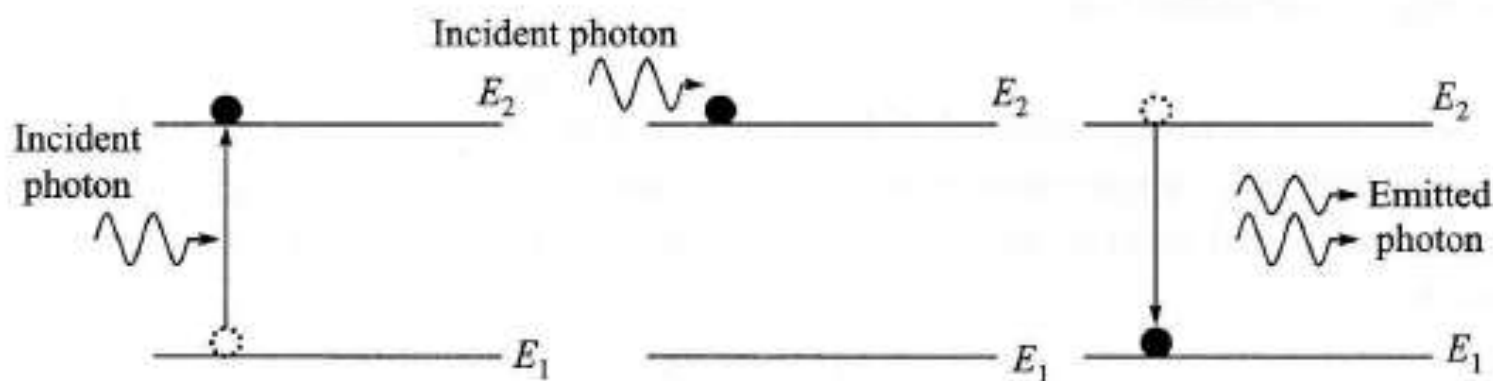


Figure 7.3 Stimulated emission.

The achievement of a significant population inversion in atomic or molecular energy states is a precondition for laser action. Electrons will normally reside in the lowest available energy state. They can be elevated to excited states by absorption, but no significant collection of electrons can be accumulated by absorption alone since both spontaneous emission and stimulated emission will bring them back down.

The process of stimulated emission is characterized by the following features:

1. The emitted photon is identical to the incident photon in all respects. It has the same frequency ν as that of incidence photon. It will be in phase with the incident photon. Both the photons travel in the same direction. They will be in the same state of polarization.
2. The process is controllable from outside.
3. The most important feature is that multiplication of photons takes place in the process. One photon induces an atom to emit a second photon, these two travelling photons can cause another stimulated emission event, giving a total of four photons, which can cause additional stimulated emissions and so on. This is the “amplification” of the laser acronym.

The probability of stimulated emission from energy State E_2 to energy State E_1 depends on the energy density of incident radiation as well as on the properties of two energy states involved. It is given by

$$(P_{21})_{\text{stimulated}} = B_{21} u(\nu)$$

Here B_{21} is the Einstein's coefficient of stimulated emissions of radiations.

The total probability of emission transition $2 \rightarrow 1$ is the sum of spontaneous and stimulated emission probabilities.

$$\begin{aligned} P_{21} &= (P_{21})_{\text{spontaneous}} + (P_{21})_{\text{stimulated}} \\ P_{21} &= A_{21} + B_{21} u(\nu) \end{aligned}$$

Difference between Spontaneous Emission and Stimulated Emission

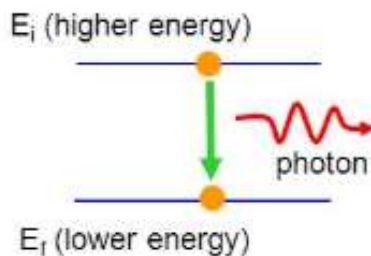
Spontaneous emission

It is a natural transition in which an atom is de-excited after the end of its life-time in the higher energy level.

The photon emitted due to spontaneous emission can move in any direction.

The probability of spontaneous emission depends only on the properties of the two energy levels between which the transition occurs.

Spontaneous emission occurs when an atom makes an unprovoked transition to a lower energy level



Spontaneous emission

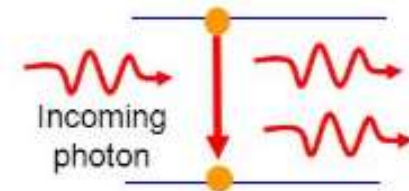
Stimulated emission

It is an artificial transition which occurs due to de-excitation of an atom before the end of its life-time in the higher energy level.

The photon emitted due to stimulated emission can move only in the direction of the incident photon.

The probability of stimulated emission depends on the properties of the two energy levels involved in the transition as well as on the energy density of incident radiation.

Stimulated emission occurs when an incoming photon induces the atom to change energy levels



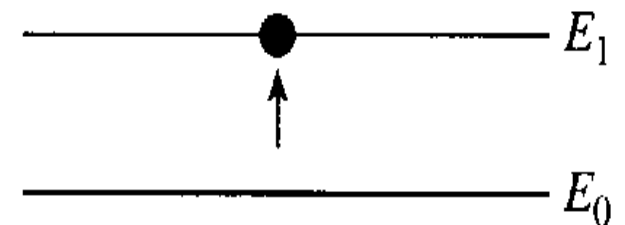
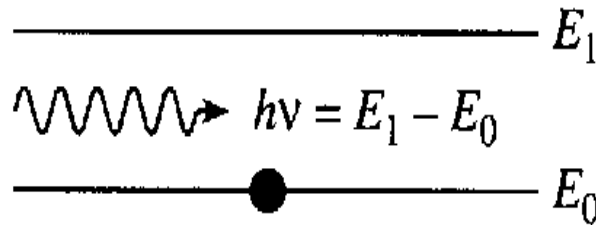
Stimulated emission

Summary of Interaction of Radiation with Matter

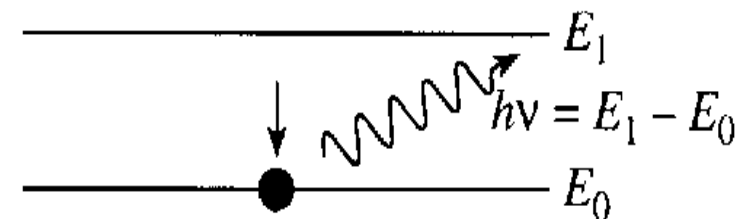
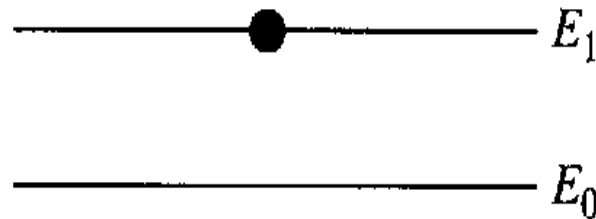
BEFORE

AFTER

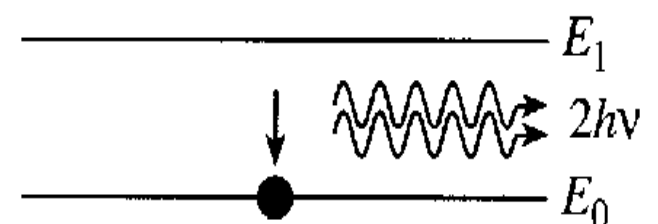
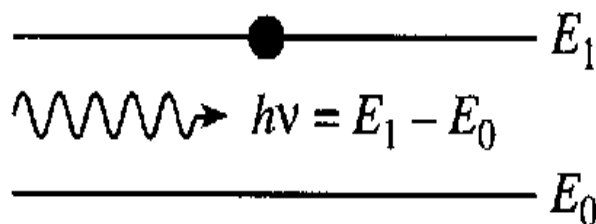
absorption



spontaneous emission



stimulated emission



Relation between Einstein's Coefficients A and B

Let us consider an assembly of atoms in thermal equilibrium at temperature T with radiation of frequency ν and energy density $u(\nu)$. Suppose N_1 and N_2 are the number of atoms in energy states 1 and 2 respectively at any instant.

The probability of absorption, that is, the number of atoms in State 1 which can absorb a photon and rise to State 2 per unit time is given by

$$N_{12} = N_1 P_{12} = N_1 B_{12} u(\nu) \quad (7.1)$$

The total probability of emission (spontaneous + stimulated) that is the number of atoms in higher State 2, which can drop to lower State 1 by emitting a photon, per unit time is given by

$$N_{21} = N_2 P_{21} = N_2 [A_{21} + B_{21} u(\nu)] \quad (7.2)$$

At thermal equilibrium, the absorption probability is equal to the total emission probability, so we have

$$N_{12} = N_{21}$$

$$N_1 P_{12} = N_2 P_{21}$$

or

$$N_1 B_{12} u(\nu) = N_2 [A_{21} + B_{21} u(\nu)]$$

or

$$N_1 B_{12} u(\nu) = N_2 A_{21} + N_2 B_{21} u(\nu)$$

or

$$N_1 B_{12} u(\nu) - N_2 B_{21} u(\nu) = N_2 A_{21}$$

or

$$u(\nu) [N_1 B_{12} - N_2 B_{21}] = N_2 A_{21}$$

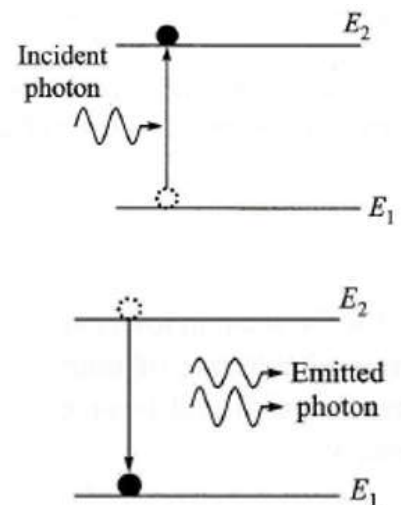
or

$$u(\nu) = \frac{N_2 A_{21}}{[N_1 B_{12} - N_2 B_{21}]}$$

or

$$u(\nu) = \frac{(A_{21}/B_{21})}{\left[\frac{N_1 B_{12}}{N_2 B_{21}} - 1 \right]}$$

or



(7.3)

or

$$u(\nu) = \frac{(A_{21}/B_{21})}{\left[\frac{N_1}{N_2} \frac{B_{12}}{B_{21}} - 1 \right]} \quad (7.3)$$

According to the Boltzmann distribution law, the number of atoms N_1 and N_2 in energy states E_1 and E_2 in thermal equilibrium at temperature T are given by

$$N_1 = N_0 e^{-\left(\frac{E_1}{K_B T}\right)} \quad (7.4)$$

and

$$N_2 = N_0 e^{-\left(\frac{E_2}{K_B T}\right)} \quad (7.5)$$

where N_0 is the total number of atoms and K_B is Boltzmann's constant.

Dividing Eq. (7.4) by Eq. (7.5), we have

$$\frac{N_1}{N_2} = \frac{e^{-\left(\frac{E_1}{K_B T}\right)}}{e^{-\left(\frac{E_2}{K_B T}\right)}} \quad \text{or} \quad \frac{N_1}{N_2} = e^{\left(\frac{E_2}{K_B T}\right) - \left(\frac{E_1}{K_B T}\right)}$$

or

$$\frac{N_1}{N_2} = e^{\left(\frac{E_2 - E_1}{K_B T}\right)}$$

or

$$\frac{N_1}{N_2} = e^{\left(\frac{h\nu}{K_B T}\right)} \quad (7.6)$$

or

$$\frac{N_1}{N_2} = e^{\left(\frac{h\nu}{K_B T}\right)} \quad (7.6)$$

where $E_2 - E_1 = h\nu$ is the energy of each photon of the radiation.

Using Eq. (7.6) in Eq. (7.3), we have

$$u(\nu) = \frac{(A_{21}/B_{21})}{\left[\frac{B_{12}}{B_{21}} e^{\left(\frac{h\nu}{K_B T}\right)} - 1\right]} \quad (7.7)$$

According to Planck's radiation formula

$$u(\nu) = \frac{8\pi h\nu^3}{c^3} \frac{1}{\left[\exp\left(\frac{h\nu}{K_B T}\right) - 1\right]} \quad (7.8)$$

Comparing Eqs. (7.7) and (7.8), we get

$$B_{12} = B_{21}$$

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

and

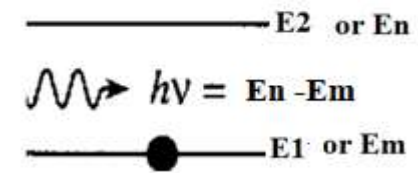
The units of Einstein coefficients may be written as

$$\frac{A_{nm}}{B_{nm}} = \frac{(J-s)(s^{-1})^3}{(ms^{-1})^3} = \frac{J-s}{m^3}$$

Physical significance of Einstein coefficient-

These equations present the relation between Einstein's coefficients. The first relation shows that the probability of stimulated absorption is equal to the probability of stimulated emission. The second relation shows that the ratio of spontaneous emission and stimulated emission is proportional to ν^3 . This indicates that the probability of spontaneous emission dominates over that of stimulated emission more and more as the energy difference between the two states increases.

Threshold Condition for Laser Action



We know that:-

The ratio of emission and absorption rates is given by

$$\frac{\text{Emission rate}}{\text{Absorption rate}} = \left[\frac{A_{nm} + B_{nm}u(\omega)}{B_{mn}u(\omega)} \right] \frac{N_n}{N_m}$$

$$\text{But } B_{mn} = B_{nm}$$

$$\Rightarrow \frac{\text{Emission rate}}{\text{Absorption rate}} = \left[1 + \frac{A_{nm}}{B_{nm}u(\omega)} \right] \frac{N_n}{N_m} \quad \dots(4.11)$$

for Absorption

$$P_{nm} = B_{nm} u(\omega)$$

for Emission

$$P_{nm} = (P_{nm})_{\text{Spontaneous}} + (P_{nm})_{\text{Stimulated}}$$

$$P_{nm} = A_{nm} + B_{nm} u(\omega)$$

The ratio of stimulated and spontaneous emission rates is given by

$$\frac{\text{Stimulated emission rate}}{\text{Spontaneous emission rate}} = \frac{B_{nm}N_n u(\omega)}{A_{nm}N_n} = \frac{B_{nm}u(\omega)}{A_{nm}} \quad \dots(4.12)$$

Therefore, for **Laser action** two conditions, given below, must be satisfied, which are known as **threshold conditions for laser action**.

(1) The emission rate must be larger than the absorption rate i.e.,

$$\left[1 + \frac{A_{nm}}{B_{nm}u(\omega)} \right] \frac{N_n}{N_m} > 1 \quad [\text{From equation (4.11)}] \quad \dots(4.13)(a)$$

(2) The probability for stimulated emission must be larger than the probability for spontaneous emission which produces incoherent radiation, i.e.,

$$\left[\frac{B_{nm}u(\omega)}{A_{nm}} \right] > 1 \quad [\text{From equation (4.12)}] \quad \dots(4.13)(b)$$

The higher probability of stimulated emission can be achieved by following methods :-

- (i) The spectral energy density of incident radiation, $u(\omega)$, must be made very larger, i.e.

$u(\omega) > \frac{A_{nm}}{B_{nm}}$ (satisfying (4.13)(b)). To increase the energy density, $u(\omega)$, the emitted radiation is made to reflect again and again between two parallel mirrors in a cavity containing active medium.

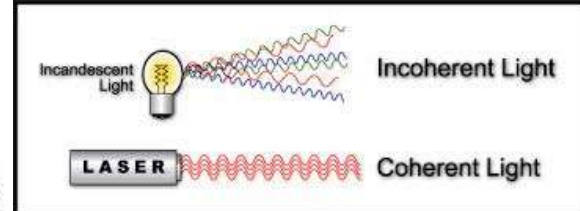
- (ii) The ratio $\frac{A_{nm}}{B_{nm}}$ must be made very small, i.e. $u(\omega) > \frac{A_{nm}}{B_{nm}}$ (satisfying (4.13)(b)). To

minimize the value of $\frac{A_{nm}}{B_{nm}}$, metastable states of higher energy is chosen for stimulated transition because transitions from metastable state by spontaneous emission are not allowed.

Using both the conditions (1) and (2) simultaneously it is clear that $\frac{N_n}{N_m} > 1 \Rightarrow N_n > N_m$.

The situation in which the number of atoms in the higher energy state exceeds that in lower energy state is known as **population inversion. This is a threshold condition for laser action.** It can be achieved by **pumping mechanism.**

Some Important Definitions



(a) Coherent and incoherent light

Coherent light is the light in which definite phase relationships exist between the waves constituting the light. The oscillations of electric field in such a light have constant amplitude at any time and the

field remains sinusoidal for infinite time as shown in Figure 7.4(a). Laser light is an example of coherent light as shown in Figure 7.5.

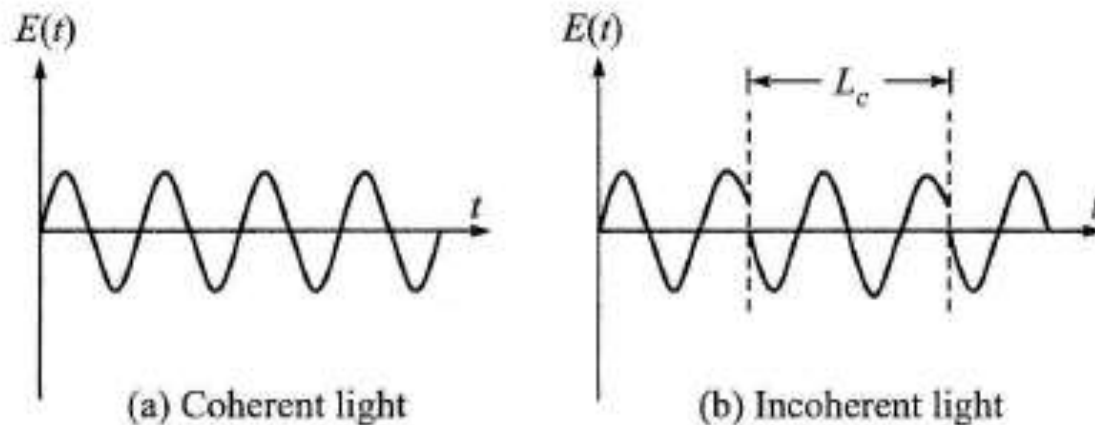
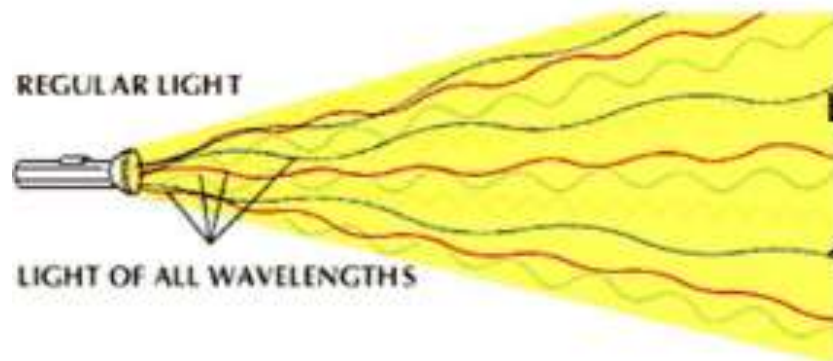


Figure 7.4 $E(t)$ vs. t plots for coherent and incoherent light.

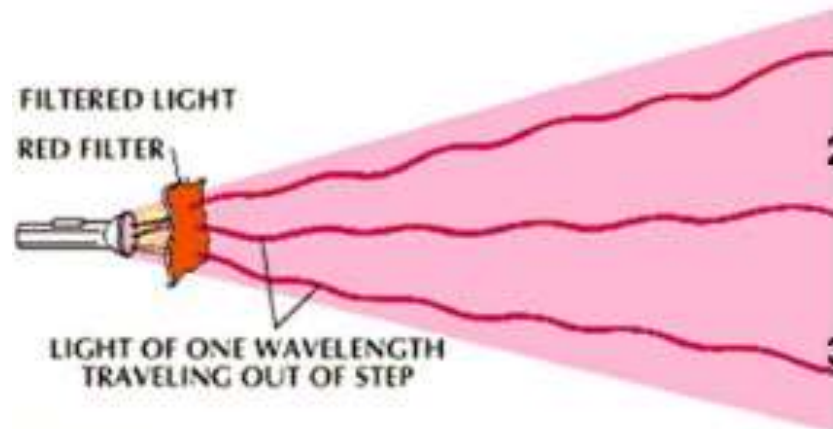
Incoherent light is the light in which there is no definite phase relationship between the waves constituting the light. The oscillations of electric field in such a light have constant amplitude at any time but the field remains sinusoidal only for a short finite time as shown in Figure 7.4(b). Ordinary light is an example of incoherent light as shown in Figure 7.5.

Coherent and Incoherent Light



Laser light has several features that are significantly different from white light.

1. White spreads out as it travels, so less light hits a surface as the distance from the light source increases. Laser light travels as a parallel beam and spreads very little.



2. White light is a jumble of colored light waves and incoherent. Laser light is monochromatic and coherent.

3. Incoherent – all the crests and troughs are NOT parallel



3. Coherent – all crests and troughs are parallel

TYPES OF COHERENCE

**(It is not in the Current Syllabus
Slide No 28 to 32)**

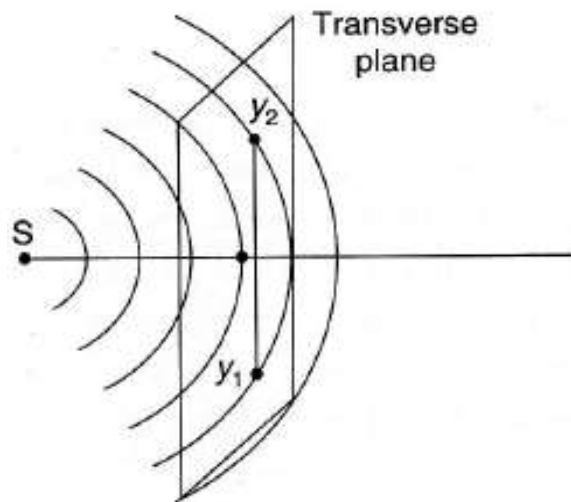
1. SPATIAL COHERENCE

2. TEMPORAL COHERENCE

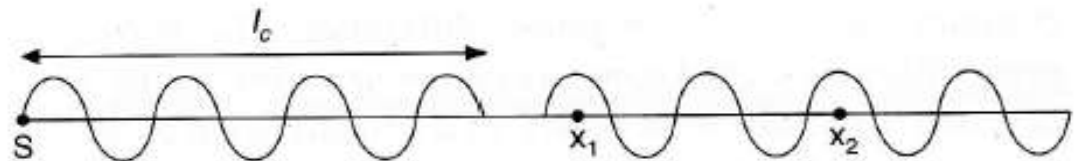
SPATIAL COHERENCE

SPATIAL COHERENCE

Consider a point source S as shown in Figure 4.1 and that waves are propagating along x direction. Then we can select two points say y_1 and y_2 along y -axis in a plane transverse to direction of propagation. Point source emits spherical waves and so, points y_1 and y_2 lie on the same or definite wavefronts and so whatever be $(y_2 - y_1)$ waves remain coherent. The point source does not exist. When S is finite, let wave phase at y_1 be ϕ_1 and at the same time wave phase at y_2 be $(\phi_1 + \Delta\phi_1)$. If $\Delta\phi_1$ remains constant and predictable then waves at y_2 and y_1 are coherent. The maximum value of Δy , the transverse separation when the phase difference $(\Delta\phi_1)$ remains constant and predictable is called *transverse coherent length* l_w . Higher is l_w , more are the waves spatially coherent.



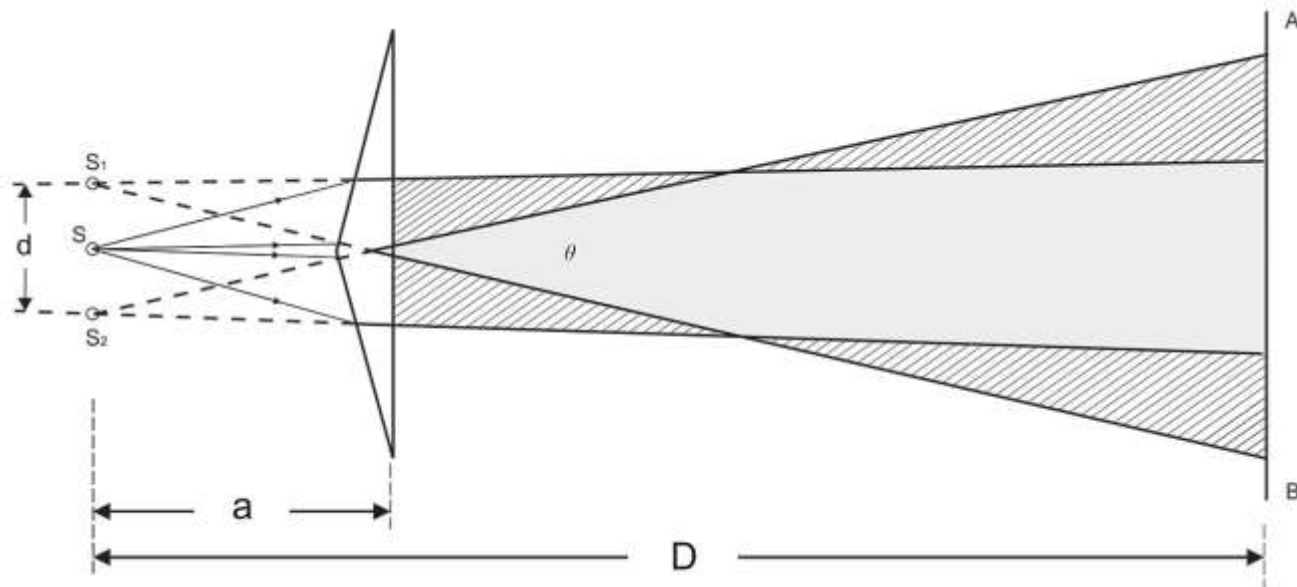
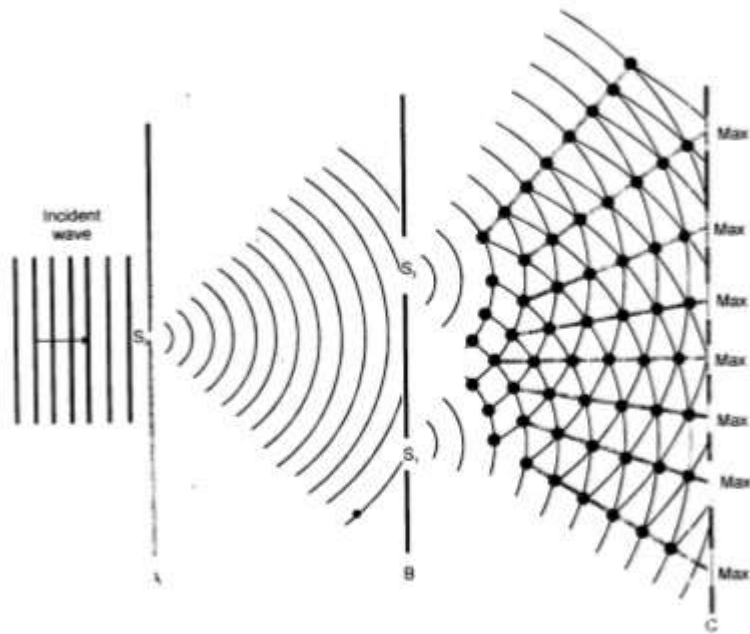
(a) Spatial coherence



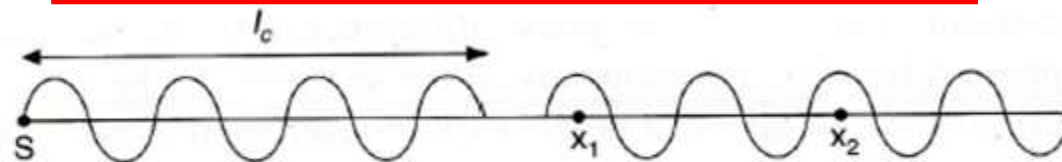
(b) Temporal coherence

- Figure 4.1** (a) When S is a point source, two points y_1 and y_2 in transverse plane maintain constant phase difference as they lie on the same or definite wavefront(s). This is not true when source has finite width.
(b) If separation of points along direction of propagation is more than wavetrain length, they have incoherent waves.

FRESNEL BIPRISM EXPERIMENT



TEMPORAL COHERENCE



(b) Temporal coherence

Temporal coherence refers to the constancy and predictability of wave phase in the direction of wave propagation. Consider two points x_1 and x_2 located in the direction of propagation and the wave phases are ϕ_2 and $(\phi_2 + \Delta\phi_2)$. If $\Delta\phi_2$ remains predictable, then the wave at x_1 and x_2 remains coherent. The maximum separation, $(x_2 - x_1)_{\max}$ along the direction of wave propagation, where the wave maintains constant and predictable phase, is called *longitudinal coherent length*.

$$(x_2 - x_1)_{\max} = l_c$$

Similarly, consider a point where wave phases at time t and subsequent time $(t + \Delta t)$ be ϕ_2 and $(\phi_2 + \Delta\phi_2)$. If $\Delta\phi_2$ remains constant and predictable, wave remains coherent for the interval Δt . The maximum time interval $(\Delta t)_{\max}$, for which the phase of the wave at a given point in the direction of wave propagation remains constant and predictable, is called *coherent time*.

$$(\Delta t)_{\max} = \tau_c$$

The higher the values of τ_c and l_c , the higher is the degree of wave coherence. The longitudinal coherent length l_c and the longitudinal coherent time τ_c are related as

$$l_c = c\tau_c \quad \text{where } c \text{ is the speed of light.}$$

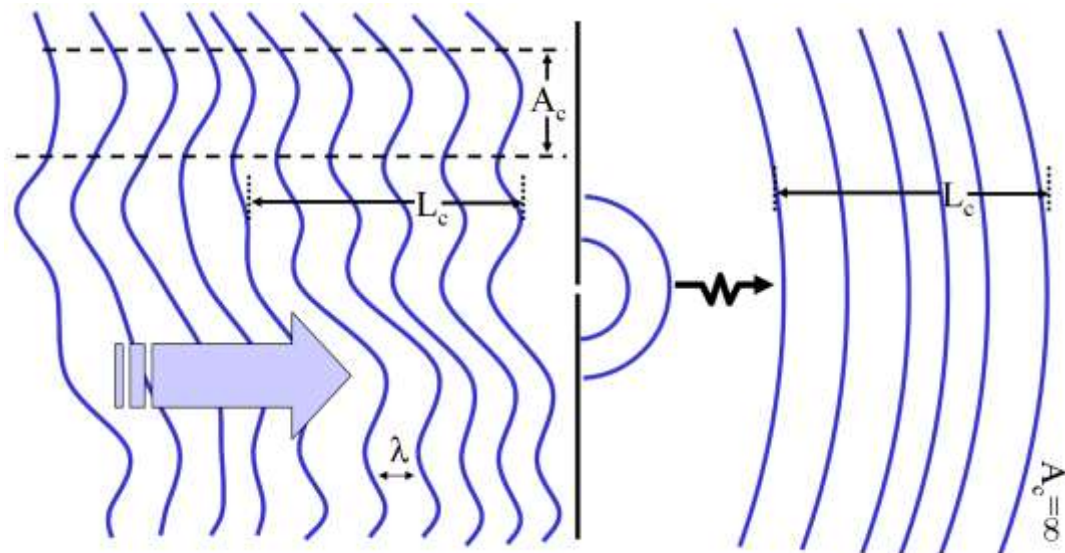
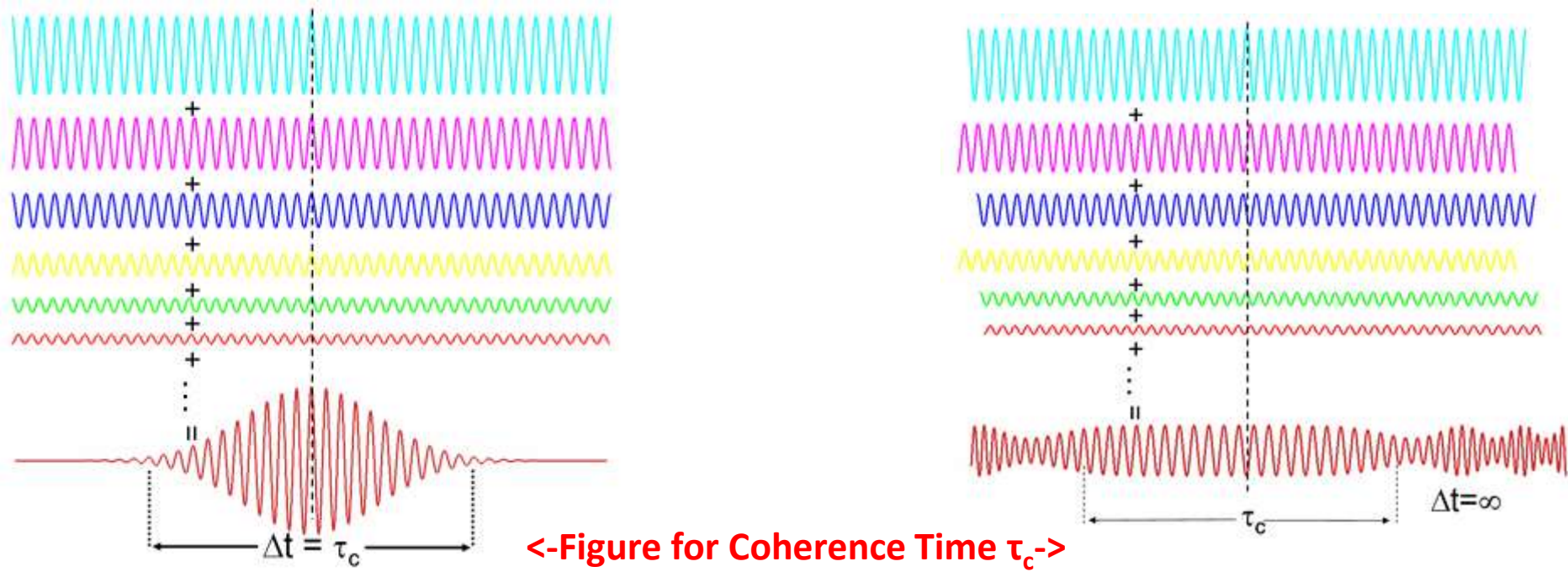
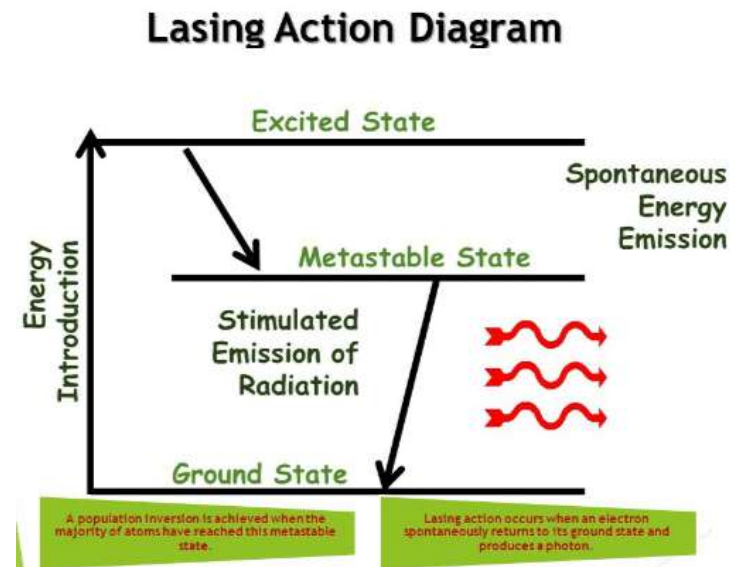


Figure for Coherence Length L_c

Metastable State



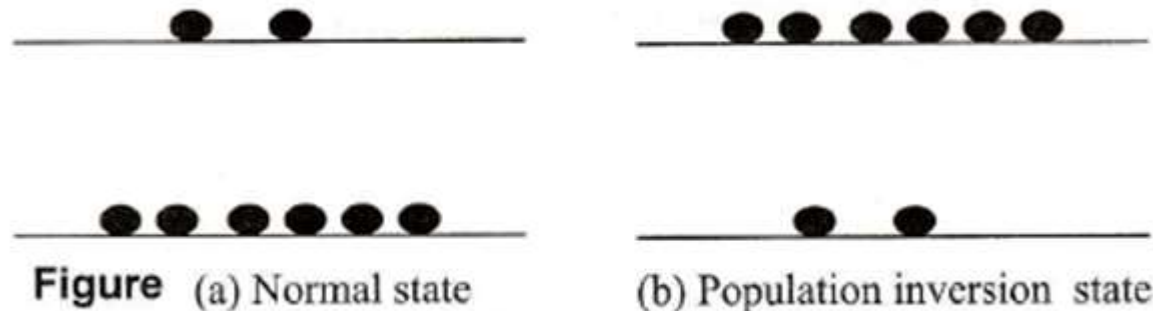
The atoms or molecules of an **active medium** have a number of energy levels **with atleast one intermediate level**. The atoms that are originally in ground state get excited to higher energy states by absorbing some external energy. Due to the short life time of an atom in higher state, atoms can not remain in the higher excited state so long and rapidly decay to lower states. As the electron drop back, some of them are being trapped in the intermediate state, called **metastable states** due to a non-radiative transition and excess energy is absorbed by the lattice as the vibrational energy of the atoms of the lattice. The life time of an atom in metastable state is of the order of 10^{-3} sec which is very much higher (about 10^5 times) than the life time (of the order of 10^{-8} sec) of the atom in the excited state. If the pumping light is intense enough, we may obtain a population inversion, i.e., more atoms in metastable state than in the ground state.

Population Inversion

- A state in which a substance has been energized, or excited to specific energy levels.
- More atoms or molecules are in a higher excited state.
- The process of producing a population inversion is called **pumping**.
- Examples:
 - by lamps of appropriate intensity
 - by electrical discharge

Population Inversion

Under normal circumstances, there is always a larger number of atoms in the lower energy state than in the excited energy state because of $N_n = N_0 e^{-E_n/kT}$ and $N_m = N_0 e^{-E_m/kT}$ and hence for $E_n > E_m$ ($\because n > m$), $N_m > N_n$. The life time of an atom in an excited state is very small (of the order of 10^{-8} sec), it makes a spontaneous emission before going to be stimulated. But the life-time of an atom in the metastable state ($\sim 10^{-3}$ sec) is very much higher than the life time ($\sim 10^{-8}$ sec) of the atom in the excited state. A passing photon produces the lasing transition from metastable state to the ground state. The above process in which number of the atoms in the metastable states is appreciably increased to the number of the atoms in the ground state is called **Population inversion** ($N_n > N_m$).



An excess of stimulated emission therefore requires a population inversion and the medium, amplify the passing photon. Thus if these conditions in the active medium are achieved, a chain reaction can be developed, resulting in **high-intensity coherent radiation**. A state of population inversion can be achieved by the **pumping mechanism**.

If the group of atoms is in thermal equilibrium, it can be shown from Maxwell–Boltzmann statistics that the ratio of the number of atoms in each state is given by the ratio of two Boltzmann distributions, the Boltzmann factor:
where T is the thermodynamic temperature of the group of atoms, and k is Boltzmann's constant.

$$\frac{N_2}{N_1} = \exp \frac{-(E_2 - E_1)}{kT},$$

Case 1. :- When $T = 0\text{K}$ [temperature = absolute zero(-273°C)]

$$N_2/N_1 = e^{(-\infty)} \Rightarrow N_2/N_1 = 0 \Rightarrow N_2=0$$

This means the population of level 2 is zero and level 1 is fully populated. At low temperatures, electrons occupy the lower level available, because there isn't enough thermal energy to raise them to the higher level.

Case 2. :- When $T = \text{infinity K}$

$$N_2/N_1 = e^{(\text{Zero})} \Rightarrow N_2/N_1 = 1 \Rightarrow N_2=N_1$$

This means the population of levels 1 and 2 are equal. Basically at very high temperatures, the populations of the levels are equal

Case 3. :- When $T = \text{Negative K}$

$$N_2/N_1 = e^{(\text{infinity})} \Rightarrow N_2/N_1 = \text{Infinity} \Rightarrow N_2>N_1$$

The term *population inversion* describes an assembly of atoms in which the majority are in energy levels above the ground state (lower energy level).

The process of achieving population inversion is known as "*pumping*" of atoms. Most commonly used methods are as follows :

1. Optical pumping (used in ruby Laser).
2. Electric discharge (used in Helium-Neon Laser).
3. Inelastic-atom-atom collisions.
4. Direct conversion (used in semi-conductor Laser).
5. Chemical reaction (used in CO_2 Laser).

COMPONENTS OF A LASER

A basic laser system has three important components as follows :

1. The active medium
2. The pumping source
3. The optical resonator

1. Active Medium

The active medium consists of a collection of atoms, molecules or ions (in solid, liquid or gaseous form) which is capable of amplifying light waves or a medium in which light gets amplified is called active medium. It is important to note that only a small fraction of a particular medium is responsible to stimulate emission and consequent amplification. These are called **active centres** and the remaining part of the medium merely supports the active centres.

A medium in which population inversion is achieved for laser action is called active medium. The medium can be solid, liquid, gas and plasma.

Under normal circumstances, there are always a large number of atoms in the lower energy state than in the excited energy state. An electromagnetic wave passing through such a collection of atoms would get attenuated. In order to have optical amplification, medium has to be kept in a state of **population inversion**, i.e. in a state in which the number of atoms in the upper energy level is greater than that in the lower energy level which may be achieved by means of the pump.

Based on the active medium and method of pumping the lasers are classified into –

- | | |
|--------------------------|--------------------|
| (i) Solid state lasers | (ii) Liquid lasers |
| (iii) Gaseous lasers | (iv) Dye lasers |
| (v) Semiconductor lasers | |

2. Pumping Source

Pumping is the mechanism of exciting atoms from the lower energy state to a higher energy state by supplying energy from an external source.

The pumping mechanism helps for obtaining such a state of population inversion between a pair of energy levels of atomic system and when we get a state of population inversion, the input light beam can get amplified by stimulated emission. This is known as optical amplification.

The most commonly used pumping mechanisms are –

- i. **Optical Pumping** : In this type of pumping, atoms are excited (i.e., population inversion is achieved) by means of an external optical source.

This type of pumping technique is adopted in solid state lasers such as ruby laser and Nd : YAG laser.

- ii. **Electrical Pumping (Direct Electron Excitation)** : In this type of pumping, the electrons are accelerated to a high velocity by a strong electric field. These moving electrons collide with the neutral gas atoms and ionise the medium. Thus, due to ionisation they get raised to a higher energy level.

This technique of pumping is adopted in gas lasers such as CO₂ laser.

- iii. **Direct Conversion** : In this type of pumping, a direct conversion of electric energy into light takes place. This technique of pumping is adopted in semiconductor laser.

In addition to the above three, other types of pumping are, inelastic collision between atoms and chemical methods which are respectively adopted in He–Ne gas laser and in dye and chemical lasers.

3. Optical Resonator

A medium with population inversion is capable of amplification, hence for that, it acts as an oscillator, a part of the output energy must be feedback into the system. Such a feedback is brought about by placing the active medium in a resonator. The resonator could just be a pair of mirrors facing each other.

Hence, optical resonator consists of a pair of mirrors facing each other in between active laser medium is placed, which provides the amplification. The mirrors provide an optical feedback and the system then acts as an oscillator. In order to produce a laser, we must collimate the stimulated emission, and this is done by properly designing a cavity or optical resonator.

A photon emitted spontaneously by an atom may cause stimulated emission. In order to sustain stimulated emission, a positive feed back of the light is required by two facing mirrors. Also spontaneous emission of photons take place in various directions and consequently incoherence occurs, again those photons moving at an appreciable angle to the walls of the cavity, or tube, will escape and be lost. Thus an intense beam of light will be available only when optical gain of the system exceeds the optical losses. Those photons emitted parallel to the axis of the cavity will reflect back and forth from end to end. Their chance of stimulating emission will now depend upon a high reflectance at the ends mirrors.

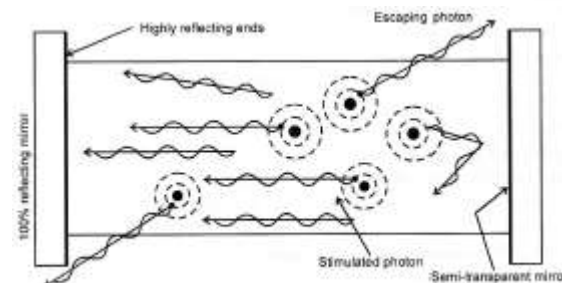


Fig. 4.2 : A Laser cavity resonator with highly reflecting ends

While propagating in the opposite direction, they de-excite more and more atoms and build up their strength. The amplified beam will move along the same path as the starting photon and undergoes multiple reflection at the mirror and gains in strength. Photon density in the medium enhance to satisfy conditions for laser.

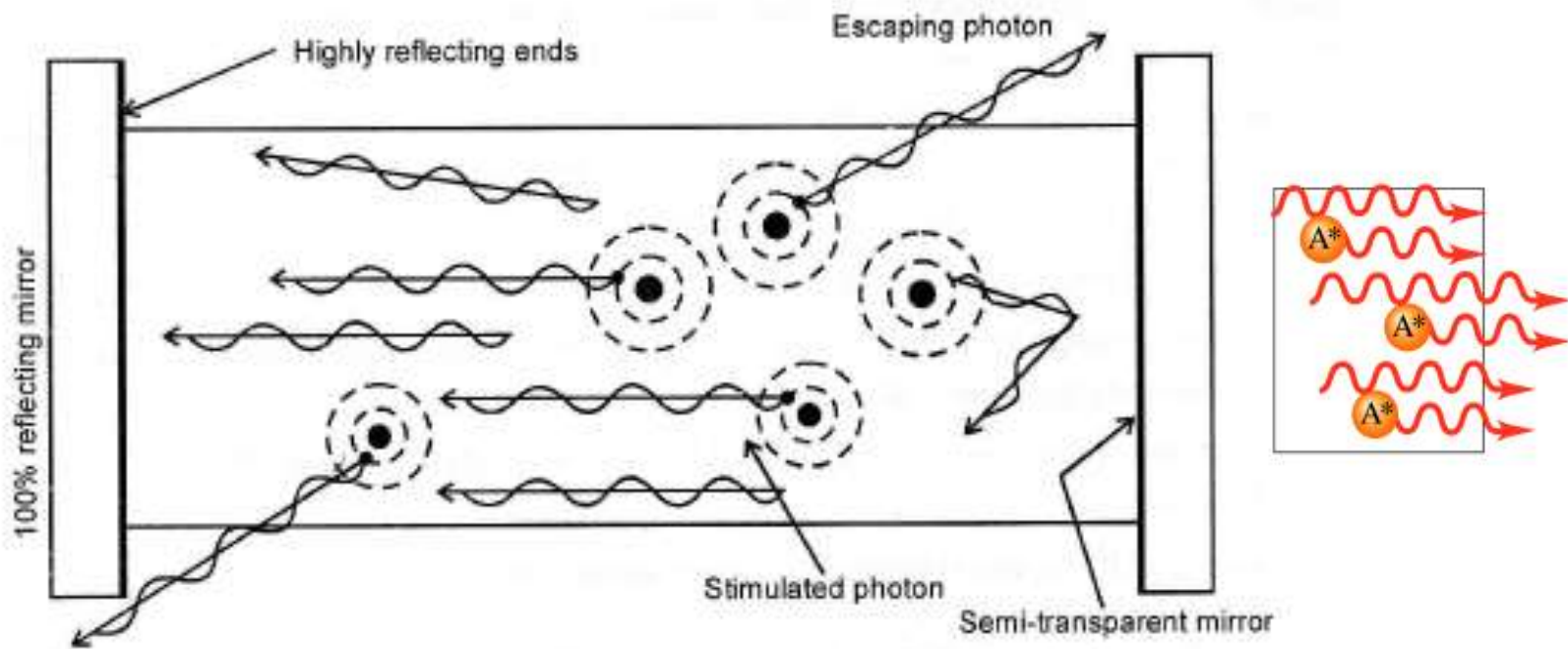


Fig. 4.2 : A Laser cavity resonator with highly reflecting ends

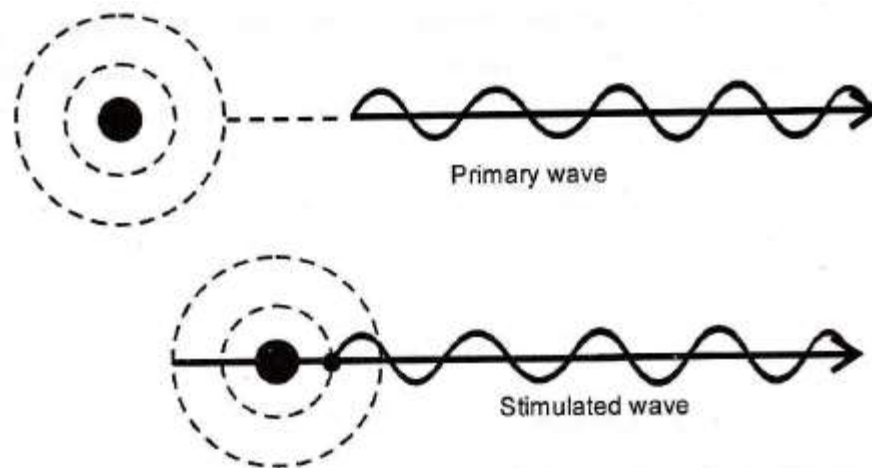


Fig. 4.3(b) : Coherent Waves

Principal of Laser

Let us consider an assembly of atoms of some kind that have metastable states of excitation energy $h\nu$. The important requirement to the laser action is that there should be more atoms in the metastable state than in the ground state. If this condition is achieved, there will be more stimulated (induced) emissions from atoms in the metastable state than induced absorptions by atoms in the ground state. The step-by-step process that takes place in laser action are :

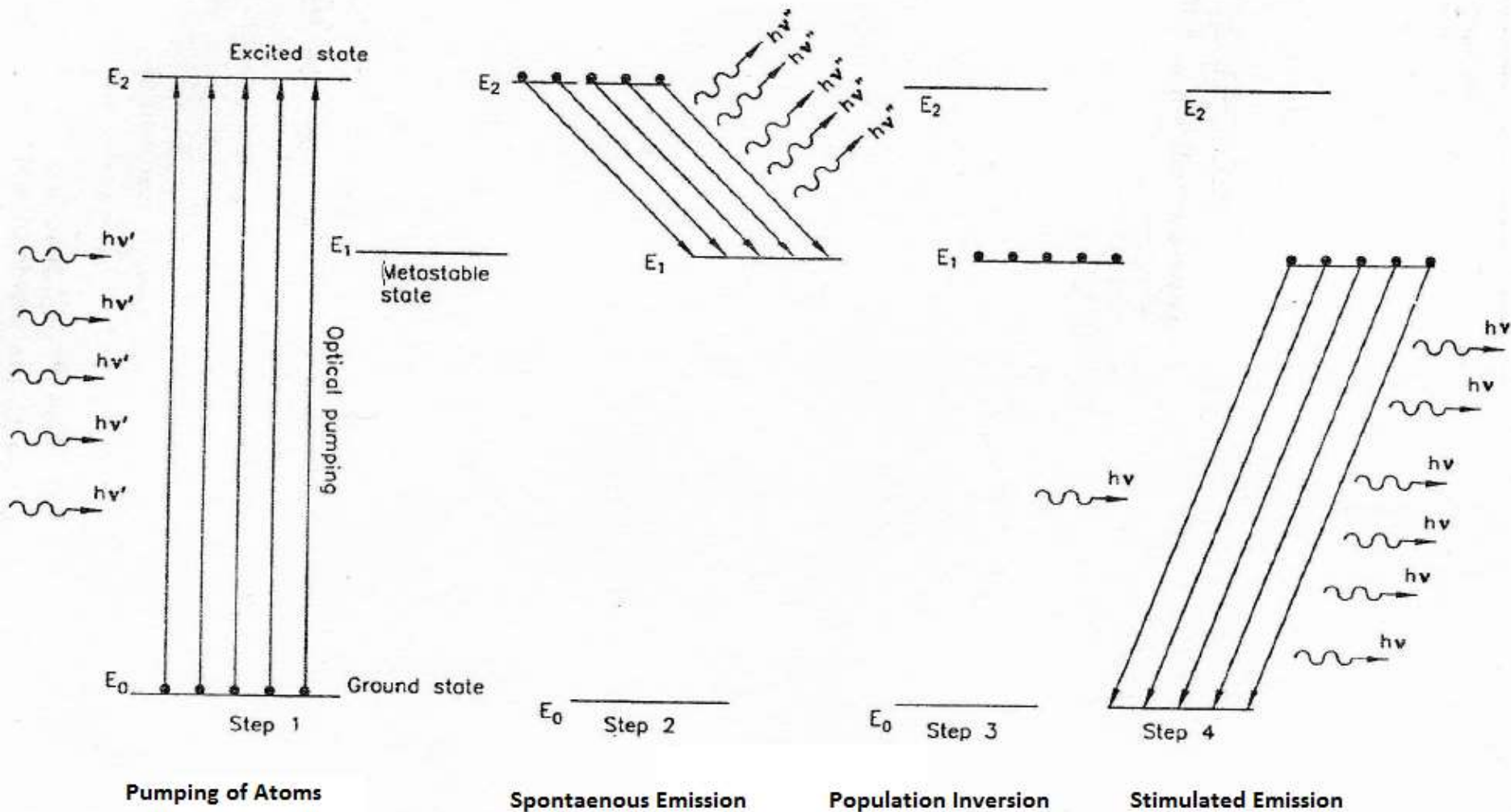
Step 1. *Atoms in ground state are pumped to excited state E_2 by photons of energy $h\nu' = E_2 - E_0$, where ν' is the frequency of incident radiation and E_2, E_0 are the energy of excited state and ground state respectively Fig. 10.4 (a).*

Step 2. *Rapid transition of metastable state E_1 by spontaneous emission of photons of energy $h\nu'' = E_2 - E_1$, where ν'' is the frequency of emitted radiation and E_1 is the energy of metastable state Fig. 10.4 (b).*

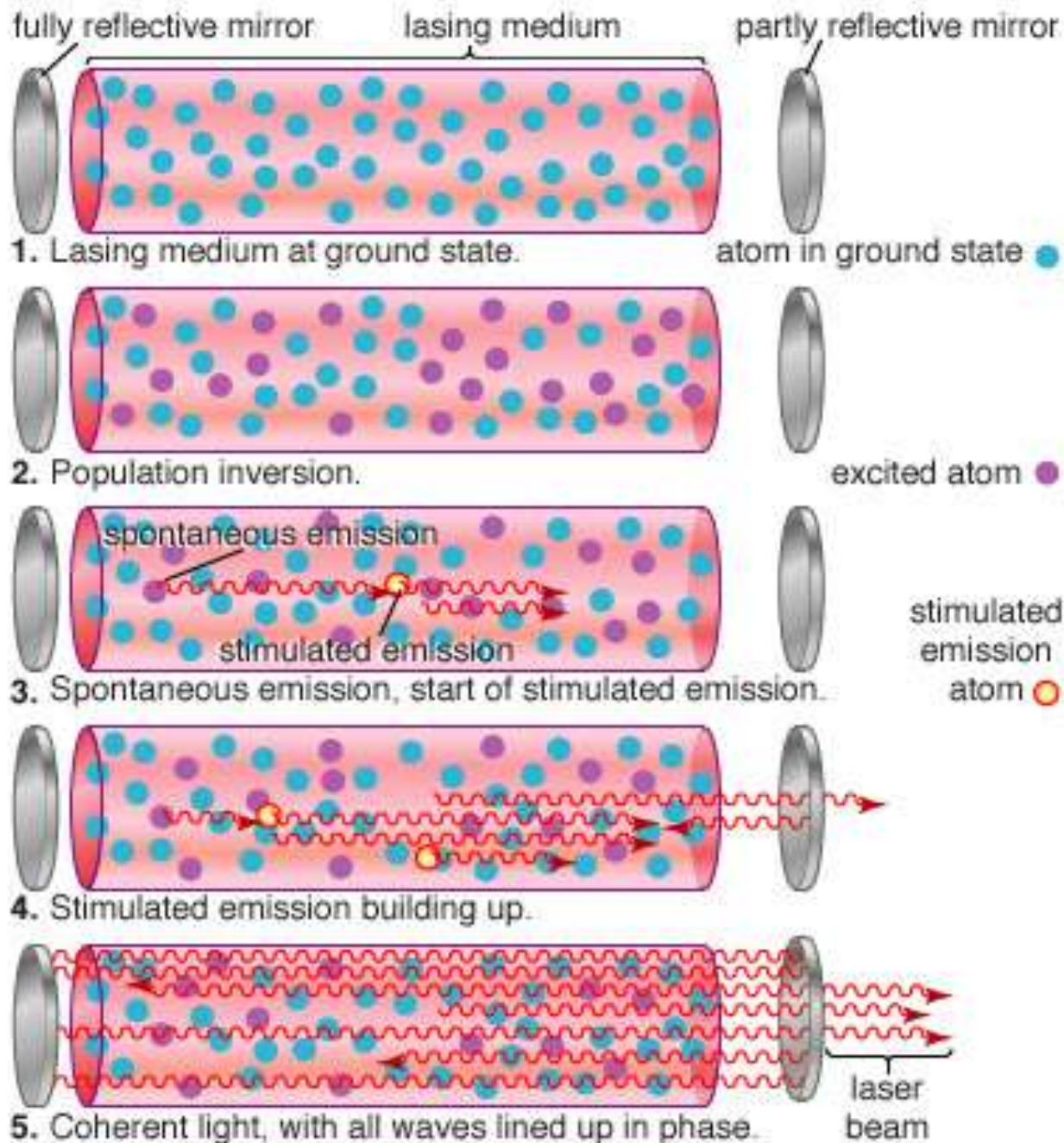
Step 3. *By this way metastable states are occupied in many atoms Fig. 10.4 (c).*

Step 4. *Induced emission occurs when photons of energy $h\nu = E_1 - E_0$ are incident with the secondary photons themselves stimulating further transitions to produce an avalanche of coherent beam Fig. 10.4 (d). (on next page).*

10.4.Principal of Laser



How a Laser Works / Basic Laser Action



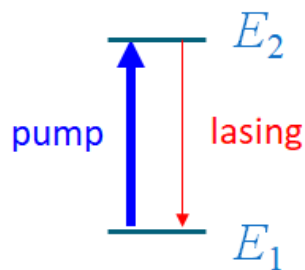
Link for Virtual Lab for How a Laser Works / Basic Laser Action :-

<https://www.youtube.com/watch?v=1LmcUaWuYao>

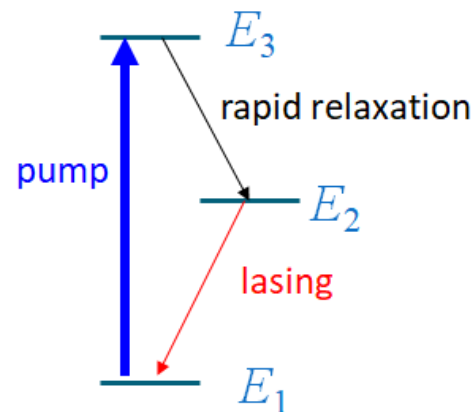
<https://phet.colorado.edu/sims/cheerpi/lasers/latest/lasers.html?simulation=lasers>

Need of Three Levels for LASER Action

Suppose there are only two energy levels, a metastable state $h\nu$ above the ground state. The more photons of frequency ν we pump into the assembly of atoms, the more upward transitions there will be from the ground state to the metastable state. However, at the same time the pumping will induce (stimulate) downward transition from the metastable state to the ground state. When half the atoms are in each state, *the rate of stimulated emissions will equal the rate of induced absorptions*, so the assembly cannot even have more than half its atoms in the metastable state. Hence a condition of population inversion will not achieve because ($N_1 = N_2$) and therefore in this situation stimulated emission will not take place and laser amplification will not occur.



Two-level system



Three-level system

Three-Level and Four-Level Laser Energy Diagrams

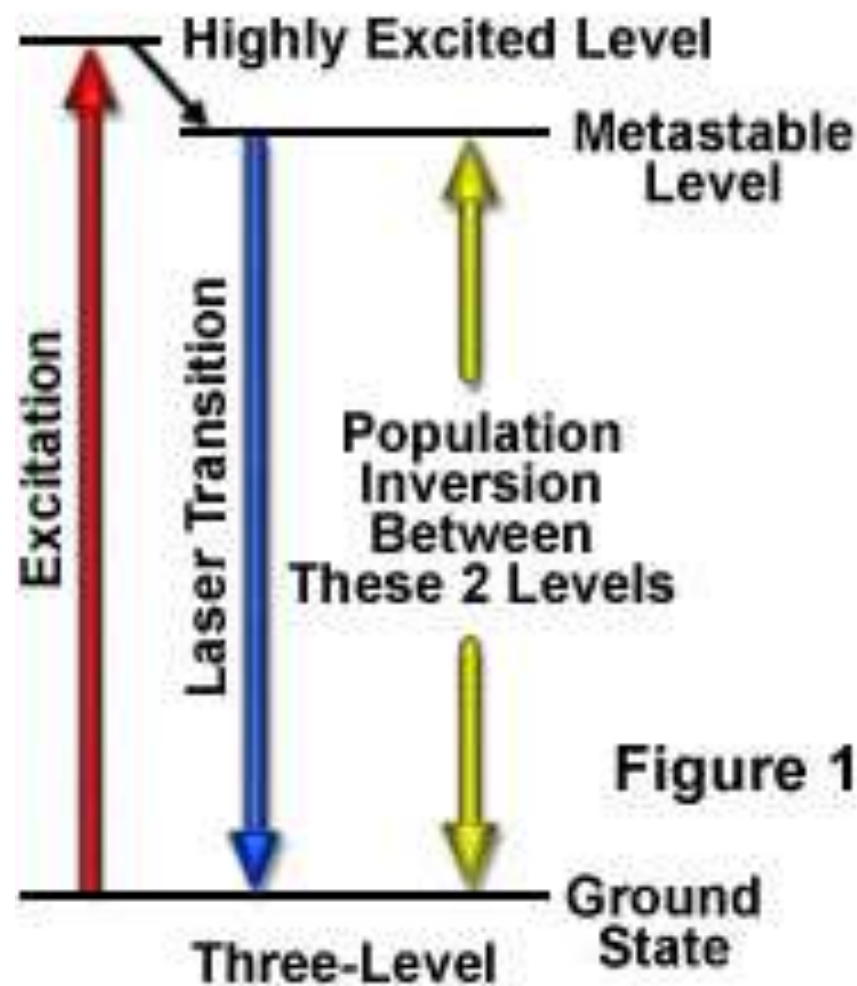
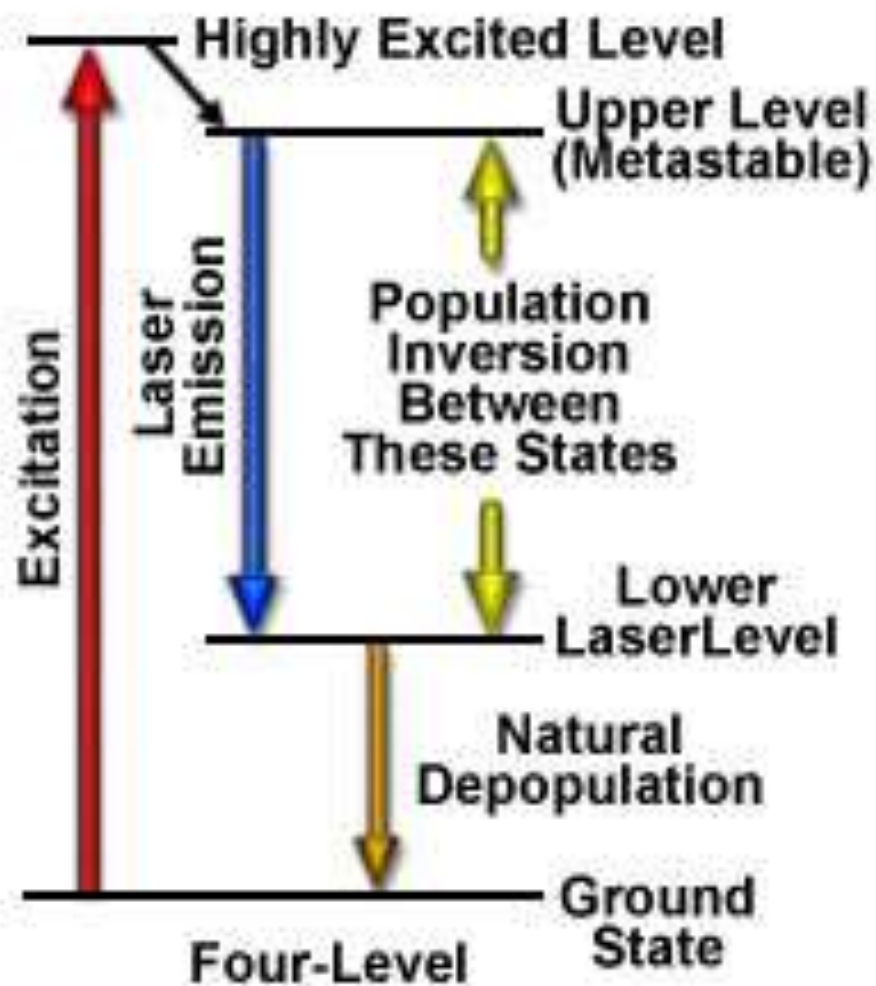
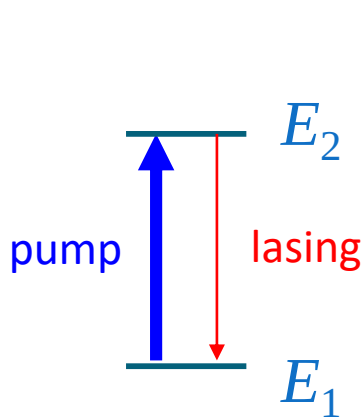


Figure 1

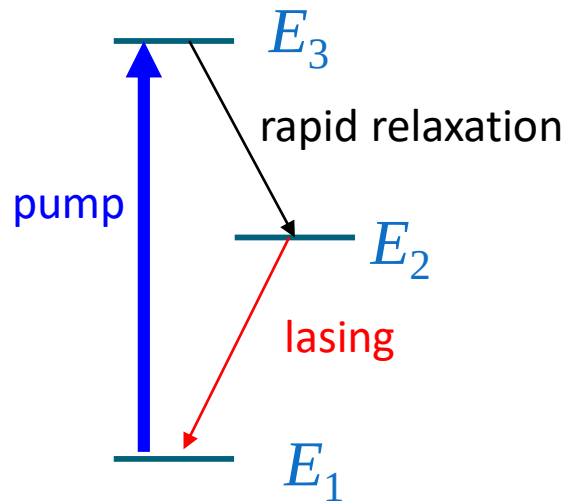


Summery of Generating a Population Inversion

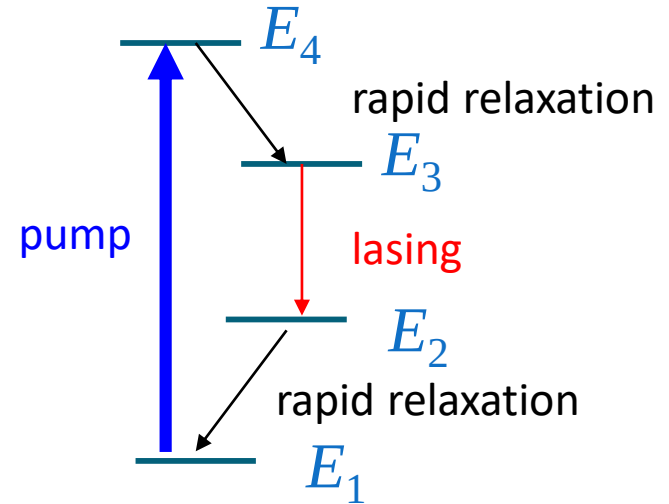
“Pumping”: delivery of energy to produce a population inversion



Two-level system



Three-level system



Four-level system

Notes:

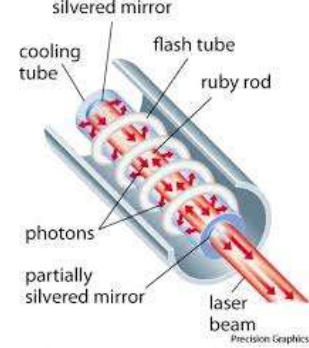
A true two-level system cannot produce a population inversion

Only E_1 is populated at thermal equilibrium ($E_2 \gg E_1$)

A three-level system must be pumped harder than a four-level system – that is, more molecules must be pumped into the excited level to produce lasing

**THANK YOU FOR YOUR
PATIENT LISTENING**

RUBY LASER



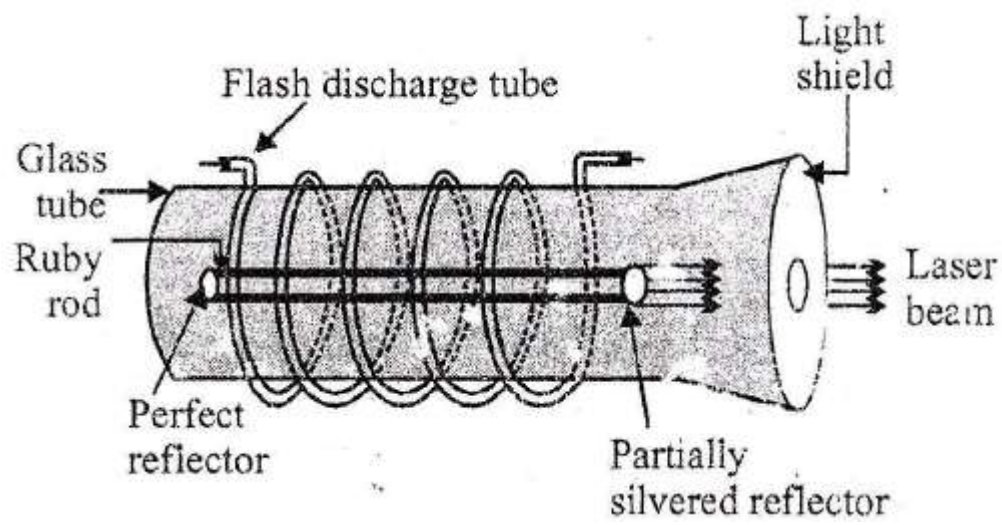
Maiman was the first who constructed Ruby laser in 1960. This device is a solid state laser and is a three level laser system. It uses optical pumping for population inversion.

Construction:-

Ruby crystal is an oxide of aluminium (Al_2O_3) with a small percentage ($\sim 0.05\%$ by weight) of chromium as impurity. In ruby crystal the Cr^{3+} ions which are responsible for the characteristic pink colour of ruby replace some of the Al^{3+} ions. The chromium ions absorb a broad band of green and yellow light in the spectral regions at $4100 \text{ \AA}$ and $5600 \text{ \AA}$ respectively.

It consists of a pink ruby cylindrical rod about 5 cm in length and 1 cm in diameter.

The ends of the rod are made optically plane and exactly parallel to each other. The ends are silver polished so that one of them is fully silvered while the other one is partially reflecting for the emergence of laser light beam.



The length of ruby rod is made equal to an integral number of half wave length of laser light.

This type of ruby rod acts as an active medium of the optical resonator cavity for laser action. The ruby rod is placed inside in a cylindrical glass tube upon which a Xenon gas discharge tube in the form of helix is wound. It is connected to a suitable power supply and made to flash repeatedly with a time interval of few milliseconds in order to pump the atoms of chromium atoms to the excited states. The cylindrical glass tube is also surrounded by another glass tube through which water is circulated to keep the ruby rod cool. The schematic diagram of ruby laser is shown in fig.

Working:-

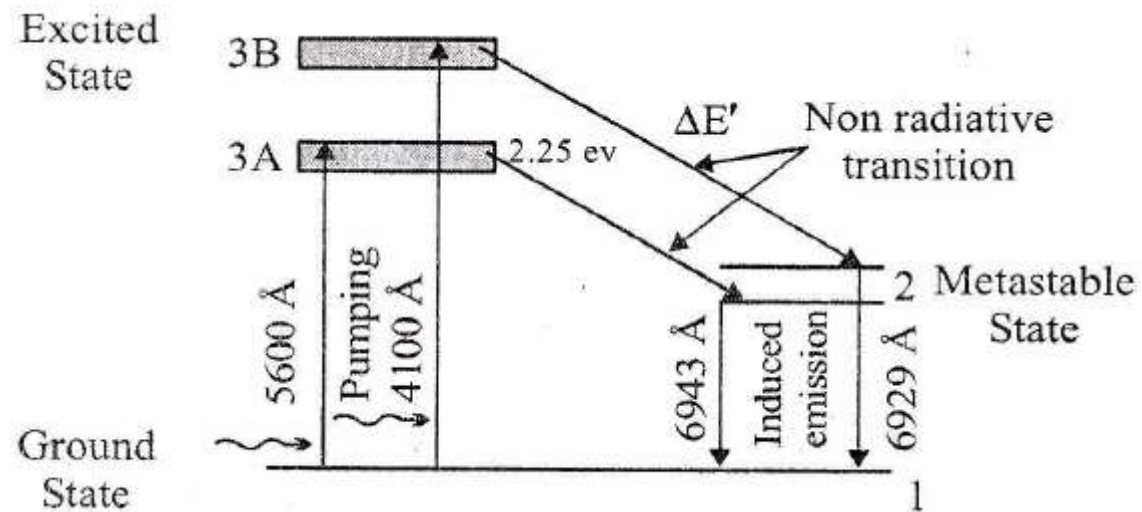


Fig.:- Energy levels of Cr^{3+}

For laser action in a ruby crystal there must be at least three energy levels of the active medium in which one of them should be metastable level in between the two levels. The chromium atoms have certain energy levels which suit the 3-level scheme for laser action. The energy levels of chromium ions have been shown in Fig. (8). The chromium ion has two broad energy levels 3A and 3B above its ground level 1 and a metastable state 2 consisting of two energy levels slightly below the energy level 3A.

When a flash of light (which lasts only for about a milli second) from the xenon flash tube falls on a ruby rod. The green and yellow light photons of definite frequencies are absorbed by the chromium ions and transitions take place to any one of excited energy states 3A or 3B. These transitions are called **pumping transitions**. The life time of ions in these levels are about 10^{-8} s.

Spontaneous transitions from the excited.....

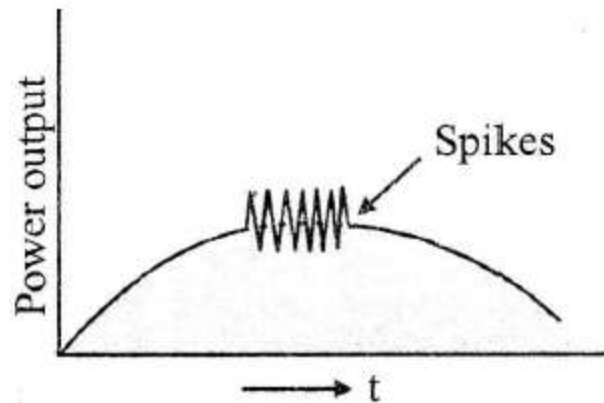
transitions from the excited states 3A and 3B make the atoms to go over to metastate 2. These transitions are non radiative in nature. The metastable states have a longer mean life of about 10^{-3} s (10^5 times greater than the usual time of excitation) so that the population density at this level grows till it exceeds the population density of ground state 1. Thus population inversion is established between the levels 2 and 1. At this stage a triggering beam of light of wavelength 6943 Å or 6929 Å will initiate laser action. Usually the wavelength 6943 Å is chosen for triggering as the probability of induced emission by this light is about 1.4 times more than that at the other wavelength.

If even one photon out of the photons produced by induced emission moves parallel to the axis of the ruby crystal, it undergoes multiple reflections between the parallel mirrors. When this photon interacts with an excited atom in the metastable state 2, it induces this excited atom to emit a fresh photon in phase with the stimulating photon. The process is repeated again and again and the photons having same phase thus multiply. The regenerative action continues and a part of the beam comes out as a red colour laser beam from the semi-transparent end of ruby crystal.

In the original set-up the Xenon flash lamp was connected to a capacitor which was charged to a few kilo volts. The energy stored in the capacitor was discharged through the xenon lamp in a few milliseconds. Thus a power a few megawatts is released and a part of this energy is absorbed for lasing. The laser output is 10-20 kW in the form of pulses.

SPIKING IN RUBY LASER

Ruby laser produces high energy pulses of coherent light of very short duration of the order of microsecond. These pulses are called spikes as shown in Fig.

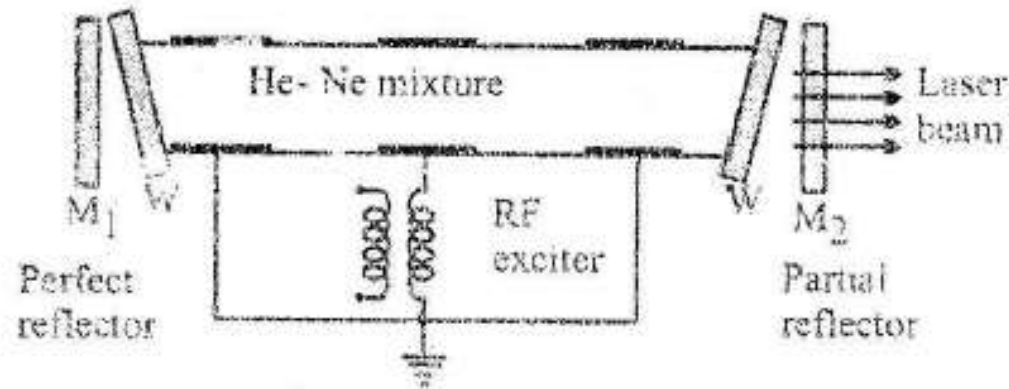


When the pulsating flash lamp is suddenly switched on, the atoms in the lower state get energy and some of these atoms transit to the metastable state. As soon as the energy given by the flash lamp reaches a threshold value (a minimum level of energy needed to produce laser action), the population inversion builds up and the laser process sets in. This in turn depletes the population of the metastable state below the threshold value. Consequently the laser action ceases for a few micro seconds. As the flash lamp continues to excite the atoms to the metastable state again laser action sets in. This lasing action repeats itself till the flash lamp power falls below the threshold value and lasing action stops.

He-Ne GAS LASER

The main defect of ruby laser is that it produces a pulsed laser beam. The development of gas-phase lasers made it possible to generate a continuous output which could not be achieved in a ruby laser. The first He-Ne laser was built by Ali Javan and his co-workers in the year 1961, which operates continuously at the visible transition in neon of Wavelength 6328 Å. In this laser, electric discharge is used as the pumping mechanism. It is a very useful device for laboratory demonstrations and experiments.

Construction



The He-Ne laser uses a mixture of helium and neon in the ratio 10 : 1 with the total Pressure of approximately 1 Torr (1mm of mercury). The mixture is kept in a thick Pyrex glass tube with an internal diameter of about 2mm and a length of about 40 cm. The windows WW' cemented at Brewster angle at the ends of the discharge tube, are of borosilicate glass and optically flat (to about one-tenth of wavelength).

The reflectors M_1M_2 are two dielectric coated spherical mirrors having a very high reflectivity at $6328 \text{ \AA}$. One mirror M_1 is a perfect reflector and the other mirror M_2 is partly transparent for the emergence of laser light. The spacing of the mirrors is equal to an integral number of half wavelength of the laser light. Pumping is achieved by an electrical discharge produced in the gas by means of electrodes outside the tube and connected to a source of high frequency alternating current. The schematic diagram of the He-Ne gas laser set up is shown in Fig.

Working:-

In He-Ne gas laser Ne atoms has metastable energy levels E_4 ($2p^5 4s$) and E_6 ($2p^5 5s$) very close to a energy states F_2 ($1s 2s$) and F_3 ($1s 2s$) of He atoms. The energy levels of He-Ne gas taking part in laser action are shown in Fig.

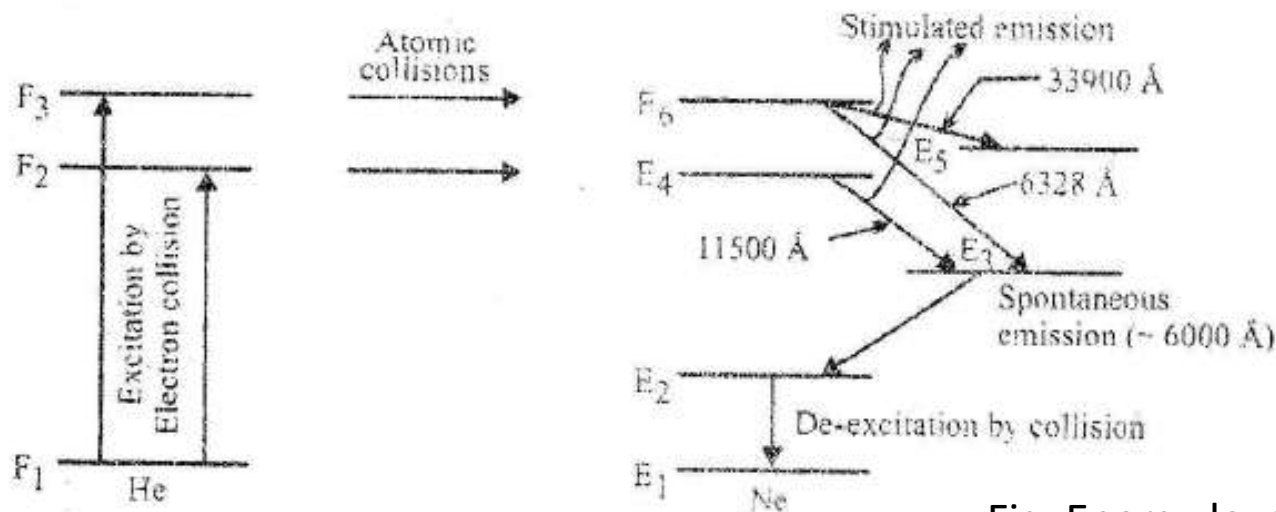
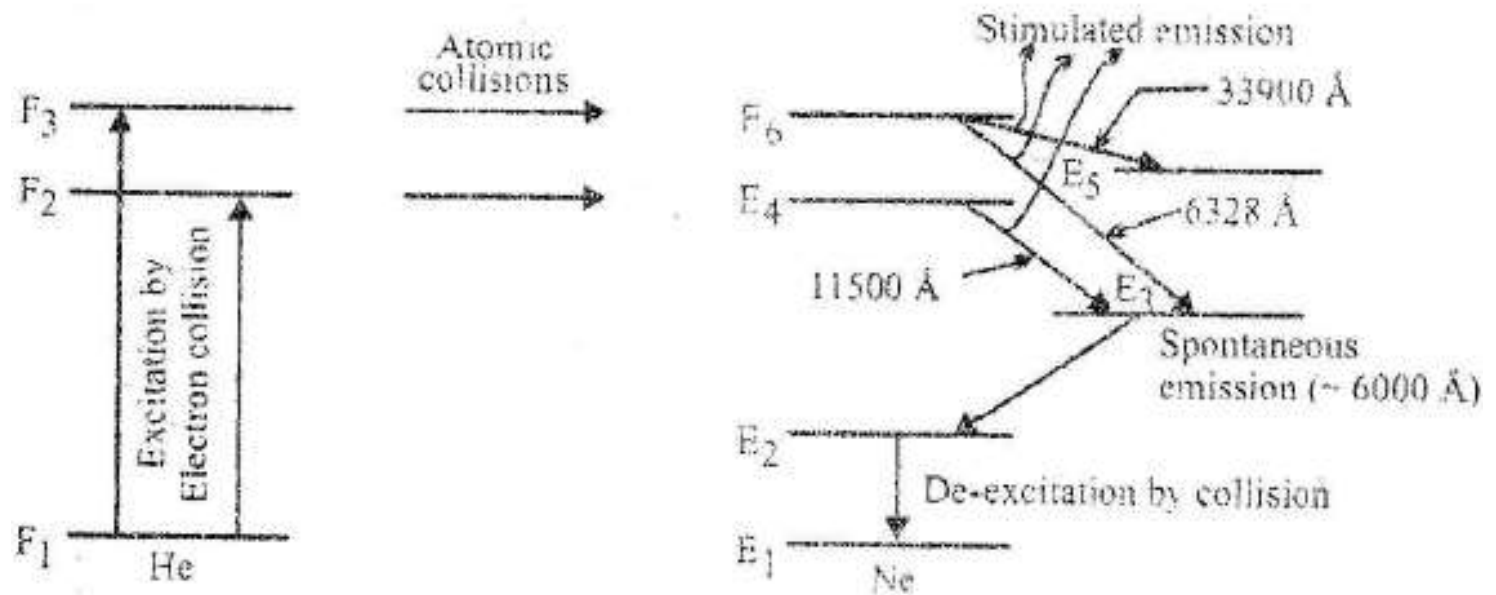


Fig. Energy levels of He-Ne gas



Energy levels of He-Ne gas

When an electric discharge is produced in He-Ne mixture by means of RF exciter, collisions between the meta stable He atoms and unexcited Ne atoms have a high probability of energy transfer to the Ne atoms. Some of the excited He atoms transfer their energy to excited state of Ne atoms by collisions. Through these collisions Ne atoms are excited to the levels E_4 and E_6 which have nearly same energy as the levels F_2 and F_3 of He. Thus the population density in the excited level of Ne will increase. When the population in E_4 and E_6 becomes much more than the population in lower levels E_3 and E_5 , population inversion is achieved. **The purpose of the He atoms is thus to achieve a population inversion in the Ne atoms.**

The transitions in Ne takes place from metastable levels E_6 and E_4 to excited states E_5 and E_3 with the emission of a $6328 \text{ \AA}$ photon in the visible region and $11500 \text{ \AA}$ and $33900 \text{ \AA}$ in the infrared region. These transition are coherent in nature. Another photon of wavelength $\sim 6000 \text{ \AA}$ is spontaneously emitted in a transition from E_3 and E_5 to a lower metastable state E_2 . This transition yields only incoherent light. The remaining excitation energy of atoms is lost in the de-excitation process by collisions with the tube and atoms reach the ground state. The high directivity of a laser beam is achieved by the parallel mirrors fixed at the end of the tube and the resonator cavity made by them. The stimulated emission is a resonant process and further the absence of such effects as crystalline imperfection, thermal distortion and scattering which are present in solid-state laser, result in getting a laser beam of wavelength $6328 \text{ \AA}$ has an extremely narrow band width (highly monochromatic). Gas lasers are capable of operating continuously and do not need cooling. The power output is however much smaller, 1-50 mW with an input of 5 to 10 W.

Comparison of Ruby laser and He-Ne gas laser

Ruby laser	He-Ne laser
1. It produces pulsating laser beam. 2. It is a three energy level system. 3. Optical pumping produces population inversion. 4. Cooling arrangement is required. 5. Laser light from this laser is less collimated and coherent. 6. It emits light of wavelength 6943 Å.	1. It produces continuous laser beam. 2. It is a four energy level system. 3. Electric discharge along with atomic collisions produces population inversion. 4. Cooling arrangement is not required 5. Laser light from this laser is highly collimated and coherent. 6. It emits light of wavelength 6328 Å.

7. Power output 10 to 20KW

8. This is Un-polarized laser

7. Power output 1 to 50W

8. This may be linearly polarized laser

Properties of a Laser

When compared to regular light, laser light has various distinct qualities that distinguish it. Those properties are

- Monochromatic
- Coherence
- Directionality
- Intense or bright

1. Monochromatic

Monochromatic light has only one color or wavelength. Ordinary light sources emit light with varying energy, frequencies, wavelengths, or colors. Laser light, on the other hand, has a single wavelength or color. The usual monochromatic light source has a bandwidth of 1000Hz. However, the bandwidth of a typical light source is 10Hz. The bandwidth is 10 to 8Hz for a sensitive laser source.

2. Coherence

If two or more waves of the same frequency are in the same phase or have a consistent phase difference, they are said to be coherent.

3. Directionality

A light ray from a normal light source goes in all directions, while laser light only travels in one. For example, the light emitted by a torchlight spreads over a one-kilometer radius. However, even though it travels less than a kilometer, laser light spreads across a few millimeters.

4. Intense or brightness

Laser light is much brighter than regular light. A 1 mW He-Ne laser is more powerful than the sun. This is due to the laser's coherence and directionality.

APPLICATIONS OF A LASER

The following are the most important laser applications:

•Medical applications of lasers

In the medical field, lasers are used for bloodless surgery, removing kidney stones, cancer detection, and treatment, correction of eye lens curvature, etc.

•Communications using lasers

Underwater communication networks, in-space communication, radars, and satellites all use optical fiber communications to transfer information across long distances with little loss.

•Industrial lasers

Lasers are used to cut glass and quartz, in the electronics sector to trim the components of Integrated Circuits (ICs), and in the automobile industry for heat treatment, etc.

• Lasers in science and technology

A laser may be used to analyze the Brownian motion of particles and count the number of atoms in a material, etc. It was demonstrated using a helium-neon laser that the velocity of light is the same in all directions.

•Military lasers

Laser range finders are used to calculate the distance between two points and high-precision hidden illuminators for nighttime surveillance. The ring laser gyroscope detects and measures small angles of rotation of moving objects.

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